Chapter 1

The Nucleus

1.1 INTRODUCTION

In this chapter, we present the concept of the nucleus, nuclear constituents and nuclear properties, such as mass, size, angular momentum, etc. To understand the concept of nucleus, it is worthwhile to review the development of various atomic models.

1.2 HISTORICAL DEVELOPMENTS

The beginning of the nuclear physics may be traced back to the studies on atomic structure started with discovery of radioactivity in 1896 by Henry Becquerel. Further, three different modes of radioactivity were observed emitting *a*-particles, *b*-particles and *g*-rays. It is well known that *a*-particles are helium nuclei, *b*-particles are either electrons or positrons and *g*-rays are high-energy electromagnetic radiations. Scattering of *a*-particles with matter revealed the existence of nucleus inside the atom. In the following paragraphs, various models are presented for establishing the structure of atom and nucleus.

Dalton's Atomic Hypothesis

The human beings have always been inquisitive to know the basic nature of the matter present in the surroundings. Breaking a piece of matter into smaller and smaller parts, search was made for its constituents, which are capable of independent existence. The atomic picture of matter is generally credited to Greeks. Around 400 B.C., the Greek philosopher Democritus postulated that all matter is made up of minute particles, which could not be broken up. Around the same times, perhaps earlier Kanada (an Indian philosopher) proposed the concept of *parmanu*, which in the similar sense,

means ultimate constituents of matter. Continuing this search, Dalton in 1803 proposed an atomic theory of matter, which has the following postulates:

1. All elements consist of discrete particles, called atoms. These atoms cannot be subdivided by any known chemical process.
2. All atoms of the same element are identical in all respects, especially in weight. Atoms of different elements differ in weight.
3. A compound is formed by the union of atoms of different elements in simple numerical proportions, for example, 1 atom of sodium combines with 1 atom of chlorine to produce one molecule of sodium chloride. Similarly, 2 atoms of hydrogen and 1 atom of oxygen combine to give a molecule of water.

Till the end of the nineteenth century, scientists did not have any clue about the structure of atom. All that they knew about the atom was that it is neutral. Works of Faraday, Maxwell and
J.J. Thomson provided the first insight into the atomic structure with the discovery of electron in 1897. It was also realized that all atoms contain electrons, which carry negative charge. Since atom is neutral, so positive charge must be present in the atom. The question arises, what kind of positive charge is present and how is positive charge arranged in the atom?

Thomson Model of Atom

Discovery of electron in the famous cathode ray experiment by J.J. Thomson provided a starting point for theories of atomic structure. Since electrons are found to carry negative charge, and atom on the whole is electrically neutral, question arose about the magnitude and distribution of positive charge inside the atom. One suggestion about the arrangement of positive charge in the atom was made by J.J. Thomson, who suggested that atoms are simply lumps of matter with positive charge and electrons are embedded in these lumps. This model is popularly known as *Plum-Pudding model*, where plums are negative electrons which are placed in the pudding of smeared positive charge. This model is also called *watermelon model*, where seeds are negative electrons. The pictorial representation is given in Figure 1.1. This model was consistent with the observed overall neutrality of the matter and explained the flow of current in matter. As it turned out later, that the real picture of the atom is quite different from Thomson's model.

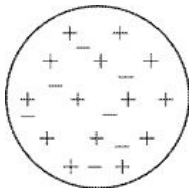


Figure 1.1 Plum-pudding model of the nucleus as given by J.J. Thomson. Plums are represented by – sign and pudding is represented by + sign.

Rutherford Model of Atom

The failure of the Thomson model was finally established in its inability to explain scattering of *a*-particles by atoms. The scattering of *a*-particles can be described in terms of the electrostatic Coulomb force between the *a*-particles and the charges which make up atoms. In the case of Thomson model, the average deflection of *a*-particles by a single atom should be very small in contrast to the experimental results. In 1911, Ernst Rutherford and his co-workers studied the way the *a*-particles got scattered when they passed through thin gold foil. They made the following observations:

1. Most of the *a*-particles went straight without getting deflected by the presence of gold foil.
2. Some *a*-particles, whose numbers were comparatively smaller, suffered small angle deviations.
3. Still smaller number of *a*-particles, about 1 in 8000, was back-scattered, i.e. through angles lying between 90□ and 180□.

Rutherford failed to explain these observations on the basis of Thomson model. From these observations, it was concluded that atom has large empty space as most of *a*-particles pass through the gold foil undeflected. So, he proposed a new picture to explain the structure of atom. He assumed that the positive charge of the atom was concentrated in a minute centre, called nucleus, instead of being distributed uniformly in whole of the atom. Thus, Rutherford transformed the static model of Thomson into a dynamic model, also known as the *Planetary Model*. He concluded that atom contains a positively charged sphere whose size was estimated to be 10^{-14} metres and he also concluded that atomic mass is concentrated in this sphere. This sphere was later on known as nucleus of the atom. There are just enough electrons around the nucleus to make atom electrically neutral. Rutherford

developed a detailed theory of the scattering of
a-particles by matter, which was tested in 1913 by Geiger and Marden. They investigated dependence of the *a*-scattering on various quantities like angle of scattering, thickness of scattering material, velocity of *a*-particles, nuclear charge, etc. The remarkable agreement between the predictions of Rutherford's theory and the experimental results established the concept of the nuclear atom. In fact, such experiments using faster *a*-particles showed deviations from the Coulomb law of scattering, which provided the first evidence of the existence of the nuclear force.

1.3 CONSTITUENTS OF THE NUCLEUS

So far, the atomic nucleus remains a vague concept. In the Rutherford model, it has been described as very small, in fact point-like, and is also supposed to contain practically all of the mass of the atom. So, in later experiments more quantitative information was sought about the size and structure of the nucleus. In the following sections, constitution of the nucleus is discussed.

As stated above, the nucleus of the atom has a very small size (its volume is 10^{-15} times the volume of the atom i.e. volume of the nucleus is of the order of 10^{-45} m^3). This compact size of the nucleus contains whole of the positive charge and practically the whole mass of the atom. Following are the two hypotheses to explain the compact structure of the nucleus:

1. Proton–electron hypothesis.
2. Proton–neutron hypothesis.

1.3.1 Proton–Electron Hypothesis

In order to explain observed properties of the nucleus, firstly it was proposed that the nucleus constitutes protons and electrons. This model is known as proton–electron model. The concept of the build-up of the nucleus in terms of the elementary constituents was based on the fact that certain atoms emit *a*- and *b*-rays, which are corpuscular in nature. As proposed by Prout, atomic weights *A* of the elements are close to integers. The fractional parts are contributed by the isotopes of the elements. The mass of the proton is approximately equal to the mass of the hydrogen atom. In fact, the hydrogen nucleus was given the name proton, which shows its importance as a fundamental constituent of nuclei of all atoms.

To account for mass of the nucleus whose atomic weight is close to integer A, called the mass number, it is necessary to assume that nucleus contains A protons. But if this was the case the charge of the nucleus will be equal to A, nearly the same as atomic weight and not equal to atomic number Z. As is well known, value of Z is half or less than half of the atomic weight. To get over this difficulty, it was assumed that in addition to protons the nuclei contain $A - Z$ electrons. The presence of electrons would not contribute to the mass of nucleus but would make the charge Z as required. Thus, it was possible to consider atom made up of a nucleus containing A protons and $A - Z$ electrons surrounded by Z extra-nuclear electrons.

Success of the Proton–Electron Hypothesis

This hypothesis seems to be consistent with the emission of a- and b-particles in radioactive elements. The presence of electrons directly ensures the emission of b-particles, and emission of a-particles is assumed by the combination of 4 protons and 2 electrons in the nucleus. These a-particles may exist as such or may be formed at the instant of emission.

Failure of Proton–Electron Hypothesis

Following are the arguments, which led to the failure of proton–electron hypothesis.

1. **Spin and statistics:** The statistical nature of nuclei can be built up from rotational spectra of diatomic molecules. If the nucleus (A, Z) contains A protons and ($A - Z$) electrons, the spin of odd–odd nucleus or odd–even nucleus would not agree with experimental results. Take the case of odd–odd nucleus ^{14}N. An even number of protons (i.e. 14) produces an integral spin, while an odd number of electrons (i.e. 7) gives half-integral spin. So the total spin of ^{14}N in the proton–electron hypothesis would be half-integral spin and thus ^{14}N would be a fermion system. But the experimental results show that ^{14}N is a boson, i.e. it carries integral spin, which could not be explained in the proton–electron hypothesis.
2. **Binding energy:** As discussed later in Chapter 8, electron being a lepton cannot take part in strong nuclear interactions. Strong interactions bind the nucleons together in the nucleus. If electrons were present in

nucleus, the bound state is caused by Coulomb interaction and the binding energy is of the order of

$$E \approx -\frac{Ze^2}{r}$$

where r is the radius of the nucleus and is given by $1.2 \square 10^{-15} A^{1/3}$ m or $1.2 \square A^{1/3}$ fm (1 fm = 10^{-15} m). Negative sign means that the system is bound. Thus,

$$E = -Z\frac{e^2}{\hbar c}\frac{\hbar c}{r}$$

$$\frac{e^2}{\hbar c} = \alpha = \frac{1}{137} \quad \text{(Fine structure constant)}$$

and $\hbar = 1.055 \times 10^{-34}$ J s

and $c = 3 \times 10^8$ m/s

Thus,

$$E = -Z \times \frac{1}{137} \times \frac{1.055 \times 10^{-34} \times 3 \times 10^8}{1.2 \times 10^{-15} \times A^{1/3}} \text{ J}$$

$$\approx -\frac{1.2 \times Z}{A^{1/3}} \text{ J}$$

If $A \approx 125$ and $Z \approx 62$, then

$$E = -\frac{1.2 \times 62}{125^{1/3}}$$

$$\approx -15 \text{ MeV}$$

The de Broglie wavelength of the electrons having energy = 15 MeV would be

$$\lambda = \frac{\hbar}{p} = \frac{c\hbar}{E}$$

Here $cp \approx E$ is the energy of the electron in the nucleus

$$\lambda = \frac{3 \times 10^8 \times 1.055 \times 10^{-34}}{15 \times 1.6 \times 10^{-13}}$$

$$= 13 \times 10^{-15} \text{ m}$$

$$= 13 \text{ fm}$$

This value is much greater than the size of the nucleus. Hence, electrons cannot exist inside the nucleus as bound electrons.

3. **Nuclear magnetic moment:** Magnetic moment of an electron is one Bohr magneton $(m_B = e\hbar/2m_e)$, while that of a proton is one nuclear magneton $m_N = eh/2m_p$. Thus,
$m_B \approx 1850 \times m_N$, where $m_N/m_e \approx 1850$. If the nucleus consisted of protons and electrons, the nuclear magnetic moment should be of the order of m_B while experimentally nuclear magnetic moment is of the order of m_N. Thus, electrons do not exist inside the nucleus.

4. **Beta decay:** As discussed later, *b*-decay is a three-body process and not a two-body process. The electrons emitted in proton–electron hypothesis would have a line spectrum, i.e. all electrons carry same energy, while in actual *b*-decay, *b*-energy varies continuously. This means that in *b*-decay, *b*-particles are accompanied by a third particle. This contradicts proton–electron hypothesis.

1.3.2 Proton–Neutron Hypothesis

After the discovery of neutrons by Chadwick in 1932 through research on transmutation of nuclei by *a*-particles, Heisenberg had earlier proposed that nuclei might be composed of protons and neutrons, collectively called nucleons. The neutron carries mass slightly greater than that of the proton, but is electrically neutral. Due to no charge, neutron was hard to detect and several unsuccessful efforts were made before it was finally observed in 1932. Thus, ${}^{16}_{8}\mathrm{O}$ nucleus contains 8 protons and 16 – 8 = 8 neutrons. ${}^{16}_{8}\mathrm{O}$ nucleus is surrounded by 8 electrons to balance the nuclear charge. In general, ${}^{A}_{Z}X$ nucleus will contain Z protons and $(A - Z)$ neutrons. To balance the nuclear charge it will be surrounded by Z electrons. This model obviously avoids the failures of proton–electron hypothesis.

Following facts support the proton–neutron hypothesis:

1. **Spin:** Both protons and neutrons have spin quantum number 1/2. According to quantum mechanics, if the number of nucleons in a nucleus is even, the resultant spin will be an integral multiple of $\hbar$. And when the number of nucleons in the nucleus is odd, the spin will be half integral multiple of $\hbar$. This observation is in agreement with the experimental observations without any exception.

2. **Magnetic moment:** According to proton–neutron hypothesis, there are no electrons inside the nucleus. Hence, we do not expect the magnetic moment of the nucleus to be of the order of Bohr magneton. On the other hand, the nuclear magnetic moment is of the order of nuclear magneton. This is in agreement with the experimental values.
3. **Finite size:** Since the mass of neutron is approximately equal to the mass of proton, it is possible for the neutron to reside inside the nucleus, according to the uncertainty principle.
4. **Wave mechanical considerations:** According to the uncertainty principle

$\Delta x \Delta p \geq h$ (1.1)

where Δx and Δp are the uncertainties in the position and momentum of proton/neutron respectively. Radius of a typical nucleus of mass ~ 200 is of the order of 0.6×10^{-14} m, the uncertainty in position will be $\Delta x = 2R = 1.2 \times 10^{-14}$ m. Therefore, uncertainty in momentum Δp is given by Eq. (1.1) as

$$\Delta p \approx \frac{h}{\Delta x}$$

$$= \frac{6.6 \times 10^{-34} \text{ J s}}{1.2 \times 10^{-14} \text{ m}} = 5.5 \times 10^{-20} \text{ J s/m}$$

Let us estimate the energy E of a nucleon in the nucleus using the relativistic relation

$$E^2 = p^2c^2 + m^2c^4$$

$$= (5.5 \times 10^{-20})^2 (3 \times 10^8)^2 + (1.67 \times 10^{-27})^2 (3 \times 10^8)^4$$

$$= 2.29 \times 10^{-20}$$

which gives

$$E = 1.51 \times 10^{-10} \text{ J} \approx 945 \text{ MeV}$$

This value is slightly greater than the rest mass energy of proton, which is about
938 MeV. Hence, the kinetic energy of neutron or proton in the nucleus is of the order of few MeV (i.e. 945 – 938 ≈ 7 MeV) and it should be possible for a free proton or neutron to be confined in the nucleus.

5. **Isotopic masses:** It is possible to explain the existence of isotopes of different elements. Different isotopes of an element have same number of protons but different number of neutrons in the nucleus.

6. **Beta decay:** The process of b^--decay can be explained by the fact that a free neutron is transformed into a proton as follows:

$$n \rightarrow p + \beta^- + \bar{\nu}$$

The process of b^+-decay can be explained by conversion of a proton into a neutron as given by

$$p \square n + b^+ + n$$

In the above decays n and $\bar{\nu}$ are neutrino and antineutrino respectively.

1.3.3 Terms Associated with the Nucleus

Atomic Number (Z)

It is the number of protons present in the nucleus. For example, nitrogen has 7 protons, so Z for nitrogen is 7, Z for uranium is 92 and for hydrogen, Z is 1.

Mass Number (A)

It is the total number of protons and neutrons present in the nucleus. For example, carbon has
6 protons and 6 neutrons in its nucleus, so its mass number is 12; uranium has 92 protons and
143 neutrons, therefore, mass number of uranium is 235, ordinary hydrogen has only 1 proton in its nucleus, so its mass number is 1. It is obvious that A can never be less than Z.

Neutron Number (N)

It is the total number of neutrons present in the nucleus and is equal to $A - Z$.

A nucleus X with atomic number Z, mass number A and neutron number N is represented as ${}^{A}_{Z}X_{N}$, for example uranium with $Z = 92$, $A = 235$ and $N = 143$ is written as ${}^{235}_{92}U_{143}$.

Isotopes

Nuclei of an element having the same atomic number but different mass numbers are called isotopes of the element. There are two types of isotopes:

(i) stable and (ii) radioactive. Stable isotopes are those which do not show radioactivity, for example, ^{14}N and ^{15}N. Some radioactive isotopes are ^{40}K, ^{60}Co, etc.

Isotones

Nuclei having same N and different Z are known as isotones. The stable isotones with $N = 1$ are $^{2}_{1}H_{1}$ and $^{3}_{2}He_{1}$.

Nucleons

The term nucleons refers to protons or neutrons present inside the nucleus. Thus, the nucleus with mass number A having N neutrons and Z protons has A nucleons.

1.4 QUANTITATIVE FACTS ABOUT NUCLEUS

Here, we introduce some facts such as size, mass, density of the nucleus and charge on the nucleus. Also the energy units commonly used in nuclear physics are discussed.

1.4.1 Size

First estimates of nuclear size were made by Rutherford. According to him, a-particles with a given kinetic energy can come closer to the nucleus till they feel a force of repulsion. The kinetic energy is converted into Coulomb potential energy. Thus, kinetic energy gets reduced and potential energy continues to increase. At a certain distance d, the kinetic energy becomes zero and potential energy becomes maximum. After this the a-particle turns back and kinetic energy starts increasing and potential energy starts decreasing. The point where kinetic energy is zero is known as distance of closest approach and it is denoted by letter d and it provides a rough estimate of the nuclear size. It is well known that in a head-on collision, the minimum distance between the projectile and the target nucleus is equal to the sum of the radii of the projectile and the target nucleus. As whole of kinetic energy gets converted into potential energy at a distance d, so the balancing equation of the kinetic energy and the potential energy is

$$\frac{1}{2}mv^2 = \frac{ZZ'e^2}{4\pi\varepsilon_0 d}$$

$$d = \frac{2ZZ'e^2}{4\pi\varepsilon_0 mv^2} \tag{1.2}$$

where m is the mass, v is the velocity and Ze is the charge of a-particle. $Z\square e$ is the charge of the target nucleus and d is the distance of the closest approach. The physical meaning of d is shown in Figure 1.2.

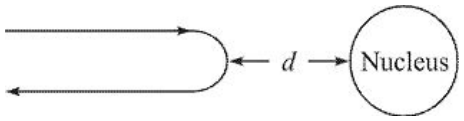


Figure 1.2 Particles being back-scattered from the nucleus, d is the distance of closed approach.

Velocity of a-particles from radon is about $1.6 \square 10^7$ m/s. The mass of a-particle is 4 times the hydrogen mass, i.e. $4 \square 1.67 \square 10^{-27}$ kg. Z is 2 and $Z\square$ is 79 for gold. e is $1.6 \square 10^{-19}$ C. The resultant value of d [from Eq. (1.2)] is $4.26 \square 10^{-14}$ m. By increasing the velocity of the a-particles one could go up to the minimum distance d equal to $3.2 \square 10^{-14}$ m for gold. For scattering of a-particles from copper nuclei the value of d is approximately $1.2 \square 10^{-14}$ m. The value of d is larger for the case of gold compared to copper. This is evident from Eq. (1.2) as d increases with $Z\square$, the atomic number of the target. In case of gold because of higher $Z\square$, d is larger compared to that of copper. It has been observed that size of the nuclei depends upon A. Presently, the accepted nuclear radius formula is

$$\boxed{r = r_0 A^{1/3}} \tag{1.3}$$

where

$$\boxed{r_0 = 1.2 \times 10^{-15} \text{ m} = 1.2 \text{ fm}} \tag{1.4}$$

$$1 \text{ fm} = 10^{-15} \text{ m}$$

Formula (1.3) is an *empirical formula*. Its derivation is not based on any theory but purely on experimental facts.

1.4.2 Mass

Nuclear masses are comparatively much smaller as compared to that of physically occurring objects. Therefore, it is generally expressed in terms of

a new unit called Atomic Mass Unit (amu). To estimate the nuclear masses we use the Avogadro's hypothesis. According to this hypothesis, a mole of a substance is that amount which has mass in grams equal to its atomic weight. For example, 1 mole of ^{12}C has 12 g. So, 1 mole of ^{12}C contains N_A = 6.02214199 □ 10^{23} atoms of ^{12}C.

Thus, the weight of 1 atom of ^{12}C is

$$\frac{12}{N_A} = \frac{12}{6.02214199 \times 10^{23}}$$
$$= 1.992646 \times 10^{-23} \text{ g}$$
$$= 1.992646 \times 10^{-26} \text{ kg}$$

For the sake of simplicity, the weight of ^{12}C atom/12 is defined as 1 Atomic Mass Unit or in short 1 amu.

Thus,

$$1 \text{ amu} = \frac{1.992646 \times 10^{-23}}{12}$$
$$= 1.660538 \times 10^{-24} \text{ g}$$
$$= 1.660538 \times 10^{-27} \text{ kg}$$

Therefore,

$$\boxed{1 \text{ amu} = 1.660538 \times 10^{-27} \text{ kg}} \quad (1.5$$

Thus, weight of 1 atom of ^{12}C is 12 amu. This is taken as standard for all the atomic weight measurements. In this unit, proton and neutron masses are given by

$$\boxed{\text{Mass of proton} = 1.0072764668 \text{ amu} = 1.67262 \times 10^{-27} \text{ kg}} \quad (1.6$$

$$\boxed{\text{Mass of neutron} = 1.0086649157 \text{ amu} = 1.674964 \times 10^{-27} \text{ kg}} \quad (1.7$$

For comparison mass of an electron in this unit

Mass of electron = 0.000549 amu = 9.109 □ 10^{-31} kg

1.4.3 Density

Density of a substance is defined as mass per unit volume.

We calculate nuclear densities for two specific cases, i.e. ^{12}C and ^{197}Au.

$$\text{Mass of } {}^{12}C = 1.992646 \times 10^{-26} \text{ kg}$$

$$\text{Radius } R \text{ of } {}^{12}C = r_0 A^{1/3}$$
$$= 1.2 \times 10^{-15} \times 12^{1/3} \text{ m}$$
$$= 2.7473 \times 10^{-15} \text{ m}$$

$$\text{Volume of } {}^{12}C = \frac{4}{3}\pi R^3$$
$$= 8.6859 \times 10^{-44} \text{ m}^3$$

Therefore,
$$\text{density of } {}^{12}C = \frac{1.992646 \times 10^{-26} \text{ kg}}{8.6859 \times 10^{-44} \text{ m}^3}$$
$$= 2.294 \times 10^{17} \text{ kg/m}^3$$

$$\text{Mass of } {}^{197}Au = 3.2707 \times 10^{-25} \text{ kg}$$

$$\text{Radius of } {}^{197}Au = r_0 A^{1/3}$$
$$= 1.2 \times 10^{-15} \times 197^{1/3} \text{ m}$$
$$= 6.9824 \times 10^{-15} \text{ m}$$

$$\text{Volume of } {}^{197}Au = \frac{4}{3}\pi r^3$$
$$= 1.4259 \times 10^{-42} \text{ m}^3$$

Therefore,
$$\text{density of } {}^{197}Au = \frac{3.2707 \times 10^{-25} \text{ kg}}{1.4259 \times 10^{-42} \text{ m}^3}$$
$$= 2.294 \times 10^{17} \text{ kg/m}^3.$$

Thus we observe that the nuclear densities are the same for carbon and gold. Or the nuclear density is constant and is of the order of 10^{17} kg/m^3. The nuclear density is extremely large. It is comparable to the density of neutron star.

In comparison to these nuclear densities, the density of carbon atom is 2.26 □ 10^3 kg/m^3 and that of gold atom is 1.92 □ 10^4 kg/m^3. Thus, nuclear densities are larger than atomic density by an order of 10^{14}.

Such a high density can be visualized by the following example. If we take 150 million cubic metres of water and compress it to 10^{-6} m^3 (or 1cc), we get value of density, which is of the order of nuclear density.

The nuclear density is independent of the mass number A. It can be explained as under:

Mass of nucleus with A nucleons = A □ m kg

where m is the mass of one nucleon.

$$\text{Volume of a nucleus with } A \text{ nucleons} = \frac{4}{3}\pi r_0^3 A$$

Therefore,

$$\text{density} = \frac{m}{\frac{4}{3}\pi r_0^3}$$

Substituting for m and r_0, we get

$$\boxed{\text{Nuclear density} = 2.294 \times 10^{17} \text{ kg/m}^3} \quad (1.8)$$

1.4.4 Energy

In nuclear physics, the unit of energy is taken to be electron volt (eV). The larger units of energy are keV and MeV. However, the SI unit of energy is joule. But SI unit of energy is not used in nuclear physics.

Electron volt (eV) is defined as the energy acquired by an electron, when it is accelerated through a potential difference of one volt.

Relation between eV and Joule

1 electron volt (1eV) = charge on 1 electron □ 1 volt

= 1.602 □ 10^{-19} C □ 1 V = 1.602 □ 10^{-19} J

i.e.

$$\boxed{1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}} \quad (1.9)$$

Therefore,

$$\boxed{1 \text{ keV} = 1.602 \times 10^{-16} \text{ J}} \quad (1.10)$$

$$\boxed{1 \text{ MeV} = 1.602 \times 10^{-13} \text{ J}} \quad (1.11)$$

In Table 1.1, commonly encountered energies are shown.

TABLE 1.1 Commonly encountered energies

Room temperature thermal energy of a molecule	0.025 eV
Visible light photons	1.5–3.5 eV
Ionization energy of atomic hydrogen	13.6 eV
Approximate energy of an electron striking a colour television screen	20,000 eV (= 20 keV)
High energy diagnostic medical X-ray photons	2,00,000 eV (= 0.2 MeV)
g-decay	0–3 MeV

b-decay	0–3 MeV
a-decay	2–10 MeV
Cosmic-ray energies	1 MeV–1000 TeV

These energy units are related to amu through mass–energy relation $E = mc^2$ as under. Using this relation, the value of 1 amu in terms of energy is

$$1 \text{ amu} = 1.660538 \times 10^{-27} \times (2.9979 \times 10^{8})^2$$
$$= 1.49239 \times 10^{-10} \text{ J}$$

$$\boxed{1 \text{ amu} = 1.49239 \times 10^{-10} \text{ J}} \quad (1.12$$

$$\boxed{1 \text{ amu} = 931.47 \text{ MeV}} \quad (1.13$$

1.4.5 Charge

Rutherford as a result of *a*-particle scattering experiments concluded that all the positive charge on an atom is confined to a tiny central region called nucleus. Later on from *a*-particle and
X-ray scattering from the atoms, it was found that the number of unit charges on the nucleus of any atom is approximately half of its atomic weight. Rutherford also concluded that proton was identical with a hydrogen ion (electron removed) that carried a single unit positive charge. Since hydrogen atom is neutral, so the charge on proton must be equal to that of electron, but of opposite sign. Similarly, *a*-particle is actually a helium nucleus, that is a helium atom minus its two electrons. Therefore, helium nucleus carries positive charge equal to $2e$, where e is the charge present on one electron. Thus, the charge on a nucleus carrying Z protons is Ze units.

1.5 BINDING ENERGY

Here we define various terms such as mass defect, packing fraction, nuclear binding energy and binding energy per nucleon. Also packing fraction has been plotted versus A and is known as *packing fraction curve*. Fission and fusion have been discussed on the basis of this curve.

1.5.1 Mass Defect

Mass of the atom is concentrated in the nucleus, which is at the centre of the atom. As discussed earlier, nucleus is constituted by neutrons and protons. It has been observed that the mass of the nucleus is always less than the sum of

the masses of all nucleons present in the nucleus. The difference in the sum of masses of all the nucleons present in the nucleus and the nuclear mass is known as the *mass defect* (Δm).

If $M'(Z, N)$ is the mass of the bare nucleus consisting of Z protons and N neutrons, the mass defect Δm is given as

$$\Delta m = ZM_p + NM_n - M'(Z, N)$$

or

$$\boxed{\Delta m = ZM_p + (A - Z)M_n - M'(Z, N)} \qquad (1.14$$

It is convenient to introduce the mass of Z atomic electrons into the right-hand side of this equation, so that it becomes

$$\boxed{\Delta m = ZM_{\mathrm{H}} + (A - Z)M_n - M(Z, N)} \qquad (1.15)$$

In this equation M_{H} and $M(Z, N)$ are the masses of neutral hydrogen and the nucleus under investigation.

Let us calculate the mass defect of deuterium ${}^{2}_{1}\mathrm{H}$. It has one proton and one neutron in its nucleus. Thus, we would expect that the mass of deuterium to be there equal to the mass of one neutron plus mass of ordinary hydrogen atom.

Mass of ${}^{1}_{1}\mathrm{H}$ atom = 1.007825 amu

Mass of neutron = 1.008665 amu

Therefore, expected mass of deuterium = 2.016490 amu

The measured mass of ${}^{2}_{1}\mathrm{H}$ is found to be 2.014102 amu. The mass difference in these masses is 0.002388 amu. Therefore, in deuterium mass defect is 0.002388 amu. This missing mass may be regarded as the mass, which would be converted into energy, if a particular atom is to be formed from the requisite number of electrons, protons and neutrons. This is also equal to the amount of energy required to break up the atom into its constituents. Therefore, mass defect is a measure of binding energy of an atom. More the mass defect, more tightly the nucleons are bound in the nucleus.

1.5.2 Packing Fraction

Mass defect does not convey much information about nuclear stability, and it is misleading to say that higher the mass defect, more tightly bound nucleons exist in the nucleus. For example,

mass defect for $^{4}_{2}\text{He}$ is 0.002604 amu (= 2.4249 MeV), while that for $^{235}_{92}\text{U}$ it is 0.04396 amu

(= 40.930 MeV). But $^{4}_{2}\text{He}$ is much more stable than $^{235}_{92}\text{U}$. The term packing fraction was introduced by Aston in 1926 which gives better information about the nuclear stability. Packing fraction is defined as

$$\text{Packing fraction } f = \frac{\text{Atomic mass} - \text{Mass number}}{\text{Mass number}} \tag{1.16}$$

or

$$f = \frac{\Delta m}{A} = \frac{\text{Mass defect}}{A} \tag{1.17}$$

The smaller the value of packing fraction the more stable is the nucleus and vice versa. A plot of packing fraction f versus mass number A is shown in Figure 1.3, from which we can draw the following conclusions:

1. For very light nuclei, like $^{2}_{1}\text{H}$, $^{3}_{1}\text{H}$, etc. packing fraction is very large, hence nucleons in these nuclei are loosely bound. In fact, out of all known stable nuclei, nucleons in $^{2}_{1}\text{H}$ are most loosely bound.
2. As mass number A increases (up to $A = 16$), the packing fraction goes on decreasing till it becomes zero for $A = 16$, i.e., for $^{16}_{8}\text{O}$ nucleus.
3. As A further increases, for nuclei with $16 < A < 180$, packing fraction becomes negative. The nucleons in these nuclei are strongly bound in the nucleus. There is a flat minimum for $60 < A < 80$. Negative packing fraction also means that in order to break these nuclei into the constituents, we must supply external energy.
4. Beyond $A > 180$, packing fraction again is positive. Thus, most of the nuclei with
   $A > 235$ are unstable.

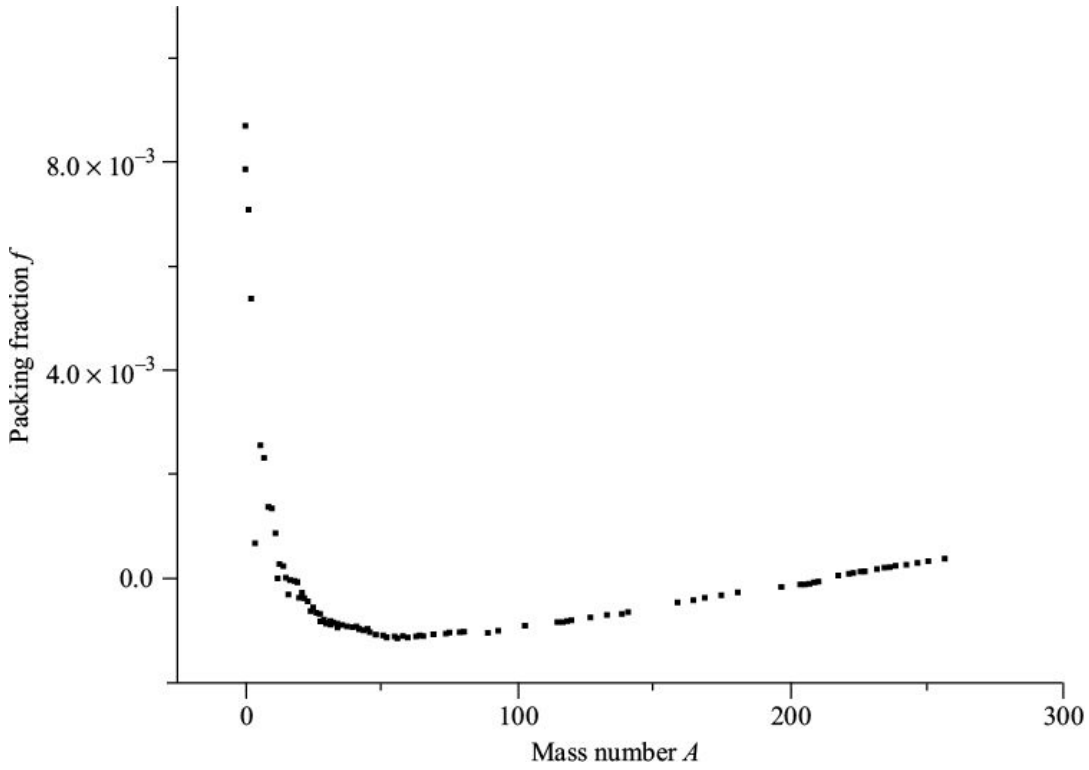


Figure 1.3 Packing fraction *f* versus mass number *A*.

1.5.3 Fusion and Fission

The phenomena of fusion and fission can also be explained on the basis of the packing fraction curve. It can be stated that each nucleus tries to reduce its packing fraction. If two light nuclei are fused together to form heavier nucleus, like

$$^{2}_{1}\mathrm{H} + {}^{3}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + n + \text{energy}$$

The packing fraction of heavier nucleus is less compared to that of the lighter nuclei. So in lighter nuclei, fusion is possible.

Similarly, if a heavy nucleus breaks into two lighter nuclei, like

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{90}_{36}\mathrm{Kr} + {}^{144}_{56}\mathrm{Ba} + {}^{1}_{0}n$$

$$^{252}_{98}\mathrm{Cf} \rightarrow {}^{106}_{41}\mathrm{Nb} + {}^{142}_{57}\mathrm{La} + 4\,{}^{1}_{0}n$$

The packing fraction of lighter nuclei is small compared to the heavier nuclei. So in heavy nuclei, fission is possible.

1.5.4 Binding Energy per Nucleon

To form an atom from requisite number of electrons, protons and neutrons, some amount of energy is required. The energy comes from the mass defect. The energy that keeps the nucleons together in a bound state is known as the binding energy of the nucleus. It can be calculated as:

The mass of constituent particles of an atom ${}^{A}_{Z}X$ is sum of masses of Z protons, $A - Z$ neutrons and Z electrons. Masses of one proton and one electron can be written as mass of one hydrogen atom (M_{H}). So masses of Z protons and Z electrons can be written as ZM_{H}. Therefore, the binding energy of ${}^{A}_{Z}X$ with mass $M(A, Z)$ is given as

$$\boxed{BE = [ZM_{\mathrm{H}} + (A - Z)M_n - M(A, Z)]c^2} \qquad (1.18$$

In driving this equation, we have neglected binding energies of the electrons in an atom, as these energies are comparatively very small. This binding energy is in amu. For converting it into MeV, we use the fact that 1 amu = 931.47 MeV. So, binding energy BE in MeV is

$$\boxed{BE = [ZM_{\mathrm{H}} + (A - Z)M_n - M(A, Z)] \times 931.47\ \mathrm{MeV}} \qquad (1.19)$$

Now binding energy per nucleon is given as

$$\boxed{\frac{BE}{A} = \frac{[ZM_{\mathrm{H}} + (A - Z)M_n - M(A, Z)] \times 931.47}{A}}\ \mathrm{MeV/nucleon} \qquad (1.20)$$

Binding energy per nucleon versus mass number A has been plotted in Figure 1.4, from which the following conclusions can be drawn:

1. Barring few exceptions, like ${}^{4}\mathrm{He}$, ${}^{12}\mathrm{C}$, ${}^{16}\mathrm{O}$, etc., the values of binding energy per nucleon lie on a smooth curve.

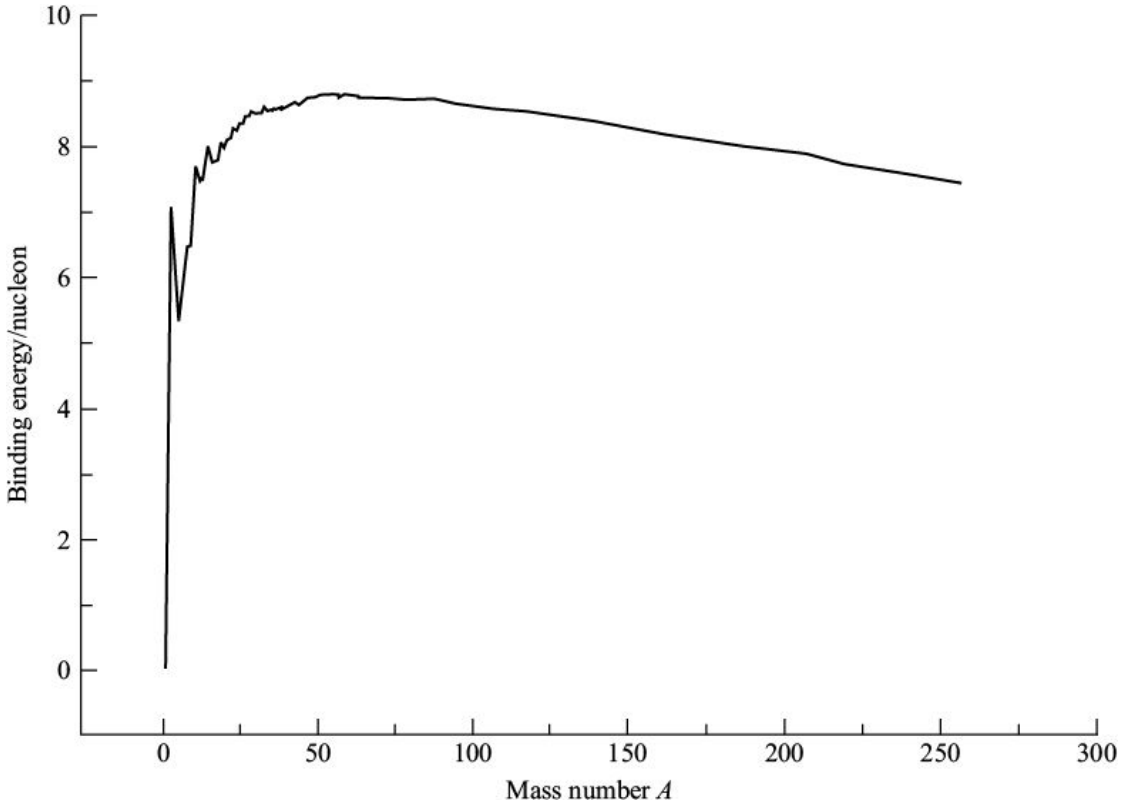


Figure 1.4 Binding energy per nucleon (in units of MeV) versus mass number A.

1. When mass number is small, i.e. $A < 12$ the binding energy per nucleon is less and it rises rapidly with increasing A.
2. Around $A = 50$, there is a flat maximum, where binding energy per nucleon is approximately 8.8 MeV. It slowly drops down to 8.4 MeV at $A = 140$. The average value of binding energy per nucleon between $A = 50$ and $A = 140$ is close to 8.5 MeV.
3. Above $A = 140$, binding energy per nucleon starts decreasing and at $A = 238$, its value is 7.6 MeV. It further reduces as A increases.
4. There are sharp peaks for ${}^{4}_{2}\mathrm{He}$, ${}^{8}_{4}\mathrm{Be}$, ${}^{12}_{6}\mathrm{C}$, ${}^{16}_{8}\mathrm{O}$, etc. nuclei. This indicates that these nuclei are more stable than the neighbouring nuclei.
5. If we take two lighter nuclei (say ${}^{2}_{1}\mathrm{H}$, binding energy ~ 1.1 MeV/n) and fuse them together to form ${}^{4}_{2}\mathrm{He}$, (binding energy ~ 7 MeV/n), there is a gain in binding energy. This indicates that fusion of lighter nuclei is energetically feasible.
6. If we take a heavy nucleus say $A = 240$ (binding energy ~ 7.6 MeV/n) and break it into two lighter nuclei of $A \sim 120$ (binding energy ~ 8.2

MeV/n), again there is a net gain in binding energy. Therefore, fission of heavy nuclei is again energetically feasible. It is left as an exercise to show that fusion of heavy nuclei is not feasible.

1.6 NUCLEAR ANGULAR MOMENTUM

Total angular momentum of the nucleus taken about its own axis, is easily measurable. But we cannot say much about the individual contributions of protons and neutrons, as their motion is very complex. One may attempt to build a vector model of the individual particles, which is similar to the vector model of the atom. To visualize the vector model, let us discuss the quantum numbers associated with the individual nucleons. The state of particular nucleon is described by its wave function, i.e. solution of its wave equation, which is characterized in terms of quantum numbers:

Principal Quantum Number (n)

Each bound particle is associated with principle quantum number n, which can take only positive integral values, $n = 1, 2, 3, \ldots$. In Coulomb field, first-order term for the total energy of the state is characterized by n, i.e. radial quantum number (Higher order terms include contributions from fine and hyperfine structures). This is not true in non-Coulomb fields such as rectangular potentials, Yukawa potentials, etc., in which nuclear particles are bound. In such potentials, the principal quantum number is a sum of radial (n) and orbital (□) quantum numbers, i.e. $n = n + $ □.

Orbital Quantum Number (□)

Orbital angular momentum quantum number (□) can take on positive integral values, i.e. 0, 1, 2, … (n – 1). The magnitude of orbital angular momentum is $\sqrt{\ell(\ell+1)}\hbar$. Various states are respectively designated as s, p, d, f, g, h, …, corresponding to □ = 0, □ = 1, □ = 2, □ = 3, □ = 4, □ = 5,…

Orbital Magnetic Quantum Number ($m_{□}$)

Orbital magnetic quantum number is the component of □ in the specified direction such as that of an applied magnetic field. It can take (2□ + 1) possible values starting from –□ to +□ differing by unity.

Spin Quantum Number (s)

Spin quantum number (s) has value 1/2 for proton, neutron and electron, which follows from Fermi–Dirac statistics and thereby obey the Pauli exclusion principle. The magnitude of spin angular momentum is $\sqrt{s(s+1)}\hbar$.

Magnetic Spin Quantum Number (m_S)

Magnetic spin quantum number is the component of s in the specified direction say that of the applied magnetic field. It can take $2s + 1$ values from $-s$ to $+s$. For particles with $s = 1/2$,
$m_S = 1/2$ and $-1/2$.

Total Angular Momentum Quantum Number ($\vec{j}$)

It is the vector sum of the orbital ($\vec{\ell}$) and spin ($\vec{s}$) angular momenta, i.e. $\vec{j} = \vec{\ell} + \vec{s}$. The magnitude of the total angular momentum is $\sqrt{j(j+1)}\hbar$. For particles with $s = 1/2$, only two values of j are permitted as $j = \square + 1/2$ and $j = \square - 1/2$. If $\square = 0$, only one value of j is allowed as $j = 1/2$.

Radial Quantum Number (n)

As stated earlier, in a non-Coulomb field, the principle quantum number n does not represent the energy of the state. In the radial wave equation solution, a radial quantum number n arises. n represents the number of radial nodes in the wave function and can have values $n = 1, 2, 3, \ldots$

Total Angular Momentum

It is the vector sum of the individual angular momenta of the constituent nucleons and is represented by J or I. It is also called nuclear spin and its magnitude is given by $\sqrt{j(j+1)}\hbar$. The value of J can be calculated in two different ways depending upon the type of coupling between angular momenta of the nucleons. The two types of coupling are:

(i) L–S coupling.
(ii) j–j coupling.

(i) *L–S* coupling: In this there is negligibly weak coupling between the orbital and spin angular momenta of the individual nucleons. Orbital angular momenta of all the nucleons couple together to give a resultant total angular momentum L. Similarly, the spin angular momenta of all the nucleons couple together to give resultant total spin angular momentum S.

$$\vec{L} = \sum_i \vec{\ell}_i$$

and

$$\vec{S} = \sum_i \vec{s}_i$$

Then the resultant L and S couple strongly to give the total nuclear spin J, i.e. $J = L + S$.

(ii) *j–j* coupling: In this scheme orbital ($\square_i$) and spin (s_i) angular momenta of individual nucleons couple together to give resultant angular momentum (j_i), or

$$\vec{j}_i = \vec{\ell}_i + \vec{s}_i$$

The different values of j_i couple together to give nuclear spin J.

$$\vec{J} = \sum_i \vec{j}_i$$

Value of J for nuclei having even number of protons and even number of neutrons is always zero. For nuclei having odd Z and even N or even Z and odd N have half integer nuclear spin (J). All odd Z and odd N nuclei have integral spin (J).

1.7 NUCLEAR MOMENTS

In this section, we discuss magnetic dipole moment of the nucleus, Bohr magneton, nuclear magneton and spin g-factor. An expression for electric quadrupole moment, which arises due to non-spherical charge distribution in the nucleus, has been derived. Prolate and oblate shapes of the nuclei have been discussed.

1.7.1 Magnetic Dipole Moment

Magnetic dipole moment of the nucleus arises due to the motion of charged particles. Orbital and spin angular momenta of protons produce magnetic field within the nucleus. This field can be described in terms of resultant magnetic dipole moment located at the centre of the nucleus.

Particle having a charge q and mass m circulates with speed v in a circular orbit of radius r. If it has a time period t, then the current i associated with the charge q is

$$i = \frac{q}{t} \qquad (1.21$$

Now t can be calculated as under

$$t = \frac{2\pi r}{v}$$

Substituting this value of t

$$i = \frac{qv}{2\pi r}$$

Now magnetic dipole moment m for current i around an area of the loop A is given as

$$m = i \square A$$

$$A = pr^2$$

Substituting the values of i and A, we have

$$\mu = \frac{qv}{2\pi r}\pi r^2$$

$$= \frac{qvr}{2} \qquad (1.22)$$

Considering the case of a single electron atom and taking q to be electronic charge e and multiplying and dividing the above equation by electron mass m, we have

$$\mu = \frac{emvr}{2m} \qquad (1.23)$$

Since mvr is the orbital angular momentum, so μ is written as

$$\mu = \frac{e\ell}{2m} \qquad (1.24)$$

where $\square = mvr$. Since $\square$ and m are vector quantities, so in vector notation, the above equation is

$$\vec{\mu} = \frac{e\vec{\ell}}{2m} \qquad (1.25)$$

However, in general, Eq. (1.25) is modified for total angular momentum

$$\vec{\mu} = \frac{e\vec{J}}{2m}g \qquad (1.26)$$

where g is a factor called gyromagnetic ratio or g-factor.

As stated earlier, J has the units of $\hbar$ so, $\frac{J}{\hbar}$ is dimensionless. Using this fact, Eq. (1.26) can be written as

$$\mu = g\mu_0 \frac{J}{\hbar} \tag{1.27}$$

where

$$\mu_0 = \frac{e\hbar}{2m}$$

The constant m_0 is called Bohr magneton and is taken as the unit to measure magnetic moments for atoms

$$\mu_B = \frac{e\hbar}{2m_e} \tag{1.28}$$

Substituting various constants, we get

$$\mu_B = \frac{1.602 \times 10^{-19}\ \text{C} \times 1.0546 \times 10^{-34}\ \text{J s}}{2 \times 9.109 \times 10^{-31}\ \text{kg}}$$

$$= 9.27 \times 10^{-24}\ \text{J/tesla}$$

For nuclei, we define magnetic moments in terms of the nuclear magneton m_N, which is defined a

$$\mu_N = \frac{e\hbar}{2m_p} \tag{1.29}$$

where m_p is the mass of proton. Substituting various constants, we get

$$\mu_N = \frac{1.602 \times 10^{-19}\ \text{C} \times 1.0546 \times 10^{-34}\ \text{J s}}{2 \times 1.67 \times 10^{-27}\ \text{kg}}$$

$$= 5.058 \times 10^{-27}\ \text{J/tesla}$$

which is smaller by a factor $\frac{m_e}{m_p}$ than the Bohr magneton.

Equation (1.25) tells us that $\vec{m} \propto \vec{\ell}$. However, the nucleus has neutrons and protons. They possess spin angular momentum in addition to orbital angular momentum. This spin gives an addition contribution to total magnetic moment of the nucleus. The spin angular momentum contribution is written as

$$\vec{\mu}_S = g_S \frac{e\hbar}{2m} \vec{s}$$
$$= g_S \times \vec{s} \times \mu_N \qquad (1.30)$$

where g_S is known as spin g-factor. Its value for proton is 5.586 and for neutron it is –3.826. Orbital angular momentum contribution to magnetic moment appears only from the protons, thus in Eq. (1.26) $g = 1$ for protons and $g = 0$ for neutrons.

Total nuclear magnetic dipole moment for the nucleus m is then given by

$$\mu = \mu_N \left[\sum_{k=1}^{A} g_k \vec{s}_k + \sum_{k=1}^{Z} g_k \vec{\ell}_k \right]$$
$$= \mu_N [g_s \vec{S} + g_e \vec{L}]$$
$$= g\mu_N \vec{J} \qquad (1.31)$$

where g is the gyromagnetic ratio of the nucleus, and $\vec{J} = \vec{L} + \vec{S}$. Magnetic dipole moment of odd Z and even N or even Z and odd N nuclei is due to unpaired single nucleon.

1.7.2 Electric Quadrupole Moment

This property arises due to non-spherical charge distribution in the nucleus. It is worthnoting that for a nucleus of spherical shape, neither dipole moment exists nor quadrupole moment exists because centre of mass of the nucleus coincides with centre of charge of the nucleus.

In non-spherical nuclei, assume that centre of mass is at the origin of the nucleus and centre of charge is not at the origin. The potential at any point P situated outside the nucleus at a distance r from the origin is given by

$$V(r) = \frac{1}{4\pi\varepsilon_o} \frac{e}{|\vec{r} - \vec{r}'|} \qquad (1.32)$$

where charge e is located at a distance r□ from the origin.
Now,

$$|r - r\square| = [r^2 + (r\square)^2 - 2rr\square \cos q]^{1/2}$$

where q is the angle between $\vec{r}$ and $\vec{r}'$, as shown in Figure 1.5. Substituting $|r - r\square|$ in Eq. (1.32), we get

$$V = \frac{1}{4\pi\varepsilon_0}\frac{e}{[r^2 + (r')^2 - 2rr'\cos\theta]^{1/2}}$$

$$= \frac{e}{4\pi\varepsilon_0 r}\left(1 + \frac{(r')^2}{r^2} - \frac{2r'}{r}\cos\theta\right)^{-1/2} \tag{1.33}$$

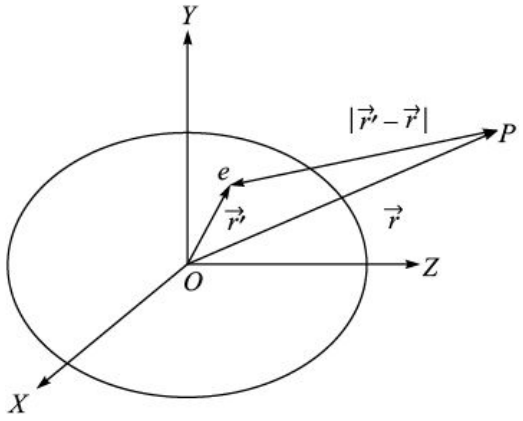


Figure 1.5 Non-spherical charge distribution. The nuclear charge e is located at a distance r□ from the origin and the potential at point P outside the nucleus at a distance r□ from the origin.

Since r□ is much smaller than r, we can carry out a binomial expansion in Eq. (1.33) and obtain

$$\left(1 + \frac{(r')^2}{r^2} - \frac{2r'}{r}\cos\theta\right)^{-1/2} = 1 + \frac{r'}{r}\cos\theta + \left(\frac{r'}{r}\right)^2\left(\frac{3}{2}\cos^2\theta - \frac{1}{2}\right) + \left(\frac{r'}{r}\right)^3\left(\frac{5}{2}\cos^3\theta - \frac{3}{2}\cos\theta\right) + \ldots$$

Substituting this expansion in Eq. (1.33), we have

$$V = \frac{e}{4\pi\varepsilon_0 r}\left[1 + \frac{r'}{r}\cos\theta + \left(\frac{r'}{r}\right)^2\left(\frac{3}{2}\cos^2\theta - \frac{1}{2}\right) + \left(\frac{r'}{r}\right)^3\left(\frac{5}{2}\cos^3\theta - \frac{3}{2}\cos\theta\right) + \ldots\right]$$

$$= \frac{e}{4\pi\varepsilon_0 r} + \frac{er'}{4\pi\varepsilon_0 r^2}\cos\theta + \frac{e(r')^2}{4\pi\varepsilon_0 r^3}\left(\frac{3}{2}\cos^2\theta - \frac{1}{2}\right) + \ldots \tag{1.34}$$

In this equation, the first term on the right-hand side represents the contribution to potential from single charge, the second term represents the contribution due to a dipole, whose electric dipole moment is

er□ cos q

and the third term represents the contribution due to electric quadrupole whose quadrupole moment Q is given as

$$e(r')^2\left(\frac{3}{2}\cos^2\theta - \frac{1}{2}\right)$$

If $Q > 0$, then nucleus is prolate in shape, as shown in Figure 1.6. In prolate shape, when two out of three principal axes are equal and the third (unequal) axis is longer than the other two axes, the nucleus will have a shape like that of a fully inflated football. More extreme examples of prolate shapes are cigar or a hot dog.



Figure 1.6 Prolate shape of the nucleus.

If $Q < 0$, then nucleus is oblate in shape, as shown in Figure 1.7. When two out of three principal axes are equal and the third (unequal) axis is shorter than the other two axes, the nucleus will have a shape like that of a pumpkin. More extreme examples of oblate shapes are burger or discus. $Q = 0$ represents spherical shape.

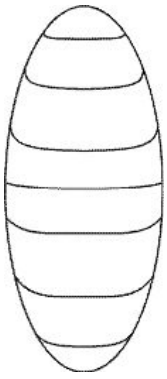


Figure 1.7 Oblate shape of the nucleus.

1.8 WAVE MECHANICAL PROPERTIES

Efforts were made to construct a theoretical model of the nucleus on the basis of classical physics, but they all failed. It was finally realized that classical physics (Newtonian mechanics, Maxwell theory of electromagnetism) can not explain atomic and subatomic phenomena. For instance, classical

electromagnetic theory, where electron, being a charged particle should radiate energy while moving around the nucleus, which would eventually collapse in the atom. Similarly, classical physics failed to explain other phenomena, like black body radiations, photoelectric effect, atomic spectra, Compton effect, etc. Initially, semiclassical–semiquantum ideas like Bohr atomic model, were developed, which could only pave the way to develop full quantum theory in two forms called wave mechanical (1927) and matrix mechanics (1925).

The fundamental idea of wave mechanics is based on the wave nature of matter. The dual wave and particle nature of matter is expressed by means of a wave function, which satisfies the following equation, first developed by Schrödinger

$$\nabla^2\psi + \frac{2m}{\hbar^2}(E - V)\psi = 0 \tag{1.35}$$

where E and m represent energy and mass of the particle moving in a potential $V(\vec{r})$. This equation successfully explained various features of atomic structure, and also introduced new concepts like parity and statistics in quantum physics. All micro-systems, also called quantum systems are expected to satisfy Schrödinger equation or some kind of wave mechanical equation. Nucleus containing nucleons is one such example, though potential experienced by nucleons within the nucleus, could be more complicated than the Coulomb potential, which worked well for the atomic spectra.

In general, any quantum system is always associated with a wave function, which depends on space–time coordinates (x, y, z, t).

Generally, y is a complex quantity but yy^* is always real. The wave function has two important properties:

1. Parity
2. Statistics

1.8.1 Parity

Under reflection, also called space-inversion, through origin x will change to $-x$, y to $-y$ and
z to $-z$. If the new wave function $y(-x, -y, -z)$ is identical to $y(x, y, z)$, then we say that the wave function has even parity. But, if the new wave function

is related to old wave function with a change of sign, i.e. $y(-x, -y, -z) = -y(x, y, z)$, then the wave function has odd parity. Alternatively, this can be represented as

$$Py(x, y, z) = y(-x, -y, -z)$$

Applying parity operator once more

$$P^2y(x, y, z) = Py(-x, -y, -z) = y(x, y, z) \quad (1.36)$$

Thus,

$$P^2 = 1$$

i.e. $P = \pm 1$ (1.37)

where P, the parity operator has values +1 or –1. The parity is a quantum concept and it does not have any classical analogue.

In hydrogen-like system, $P = (-1)^{\ell}$, where ℓ is the orbital quantum number. Depending upon ℓ, P has the value +1 or –1.

A system having even number of odd parity particles has even parity and a system having odd number of odd parity particles has odd parity. Because parity of the system is the product of individual parities.

Except for weak interactions, discussed in Chapter 8, parity is always conserved in a nuclear transformation, e.g. if a even parity excited nuclear state decays by *g*-emission, then the product of parities of *g*-ray and the daughter nucleus must be even. Similarly, in *a*-decay if the parent nucleus has even parity, then the product of parities of the daughter nucleus and the *a*-particle must be even.

1.8.2 Statistics

The concept of statistics is related to the behaviour of large number of particles. For instance, Maxwell–Boltzmann statistics works well in the classical physics, while describing the distribution of energy among the molecules of a gas. However, in quantum physics, two new forms of statistics have been developed which are the Bose–Einstein statistics and the Fermi–Dirac statistics. These statistics reflect in the behaviour of wave function describing systems of identical particles under their interchange.

Bose–Einstein Statistics

If the particles have integral spins, i.e. 0, 1, 2, … , they are known as Bosons, e.g. photons, pions, etc. The wave function of the system consisting of Bosons is symmetric when we interchange
any two identical Bosons, i.e. $y(x_1, x_2) = y(x_2, x_1)$. Similar results can be written for more than two particles.

Fermi–Dirac Statistics

The particles having half integral spins are known as Fermions. Examples are protons, neutrons, etc. All Fermions obey Pauli's exclusion principle. The wave function of a Fermions system is antisymmetric under interchange of two identical particles, i.e. $y(x_1, x_2) = -y(x_2, x_1)$.

1.9 NATURE OF NUCLEAR FORCES

The forces known in the beginning of the twentieth century—gravitational and electromagnetic Coulomb forces could not account for the stability of the nucleus. Protons carry positive charge and neutrons are neutral particles. The electrostatic force will be repulsive between two protons. So, this force will cause the nucleus to break and thus it cannot account for the binding of the nucleus. The gravitational force of attraction between two nucleons is extremely weak, it also cannot explain the binding energy of the nucleus. This force is significant only on the macroscopic scale, such as for astronomical objects.

The small size of the nucleus (~ a few fermi) and its great stability shows that the forces operating among the nucleons are short-range forces, i.e. they are effective over very short distances only (~ few fermi). This short-range character distinguishes the nuclear forces from gravitational and electromagnetic forces, which act over large distances. If the forces operating among the nucleons were long range, there would be interactions among distant nucleons in the nucleus, as well as between those in close proximity; the total binding energy would increase roughly as A^2. However, the actual binding energy is approximately proportional to A. This characteristics of the forces is called saturation property. Each nucleon attracts those in its immediate vicinity, but is relatively indifferent to the others. This new force, called nuclear force was introduced to account for the stability of the nucleus. This force was assumed to be much stronger than the repulsive Coulomb

force between two protons. This force is attractive between 2 protons, 2 neutrons or 1 proton and 1 neutron. Later on, these nuclear forces were found to be highly complex. Some important characteristics of nuclear forces are:

Nuclear Forces are Charge-Independent

The force between *n–n* is same as that between *p–p* and between *n–p*. This is termed as charge-independence of nuclear forces. This is clear from the formation of mirror nuclei. For example, ${}^{3}_{1}H$ and ${}^{3}_{2}He$ are mirror nuclei. The former contains 1 proton and 2 neutrons, while the later contains 2 protons and 1 neutron. The binding energy of ${}^{3}_{1}H$ is 8.5 MeV and that of ${}^{3}_{2}He$ is

7.7 MeV. The three types of forces (*n–n*, *p–p*, *n–p*) are equal in magnitude, the difference of

0.8 MeV in ${}^{3}_{1}H$ and ${}^{3}_{2}He$ is due to repulsive Coulomb energy between proton–proton in ${}^{3}_{2}He$ nuclide.

Nuclear Forces are Very Strong in Nature

They are much stronger than electromagnetic and gravitational forces. They are stronger by a factor of 137 from the electromagnetic force and are stronger by a factor of 10^{40} from gravitational forces.

Nuclear Forces have Short Range

Nuclear forces are short-range forces and the range is of the order of nuclear dimensions. If the range of the nuclear forces is larger than the dimensions of the nucleus, they would have definitely affected molecular interactions, but no such effect has ever been observed in molecular structure.

Nuclear Forces are Repulsive at Very Short Distances (*~0.5* fm)

If this was not so, the nucleons inside the nucleus would collapse to the size of nuclear force range.

Nuclear Forces are Saturated

The range of nuclear forces is limited. Saturation of nuclear forces implies that their interaction is only with the nearest neighbours. The effect of

saturation property of nuclear forces is that the binding energy is proportional to A.

Nuclear Forces are Non-Central

Electrostatic and gravitational forces are central forces, i.e. these forces act along the line joining the two particles while nuclear forces depend not only on the distance between the nucleons, but also on the angle that their spins make with the line joining the two nucleons.

Nuclear Forces are Spin and Angular Momentum-Dependent

Two nucleons having parallel spin have stronger force between them than those having anti-parallel spin. This force depends on the angles between the spin and orbital angular momentum vectors.

Nuclear Forces are Velocity-Dependent

Nuclear forces depend upon the velocity of the nucleons.

Nuclear Forces are Exchange Forces

It means that the force is produced by exchange of force carrier. It is described in detail in the next section.

1.10 YUKAWA THEORY OF NUCLEAR FORCES

Yukawa in 1935 proposed a theory to explain the nature of nuclear forces. Yukawa proposed that when two nucleons are interacting, there is a field associated with this system and the cause of the force between the two nucleons is an exchange of a quantum between them. A quantum is emitted by one nucleon and is absorbed by another.

This assumption is similar to the case of Coulomb force between two charged particles. According to quantum field theory, the Coulomb interaction between the two is explained by saying that one charged particle emits a virtual photon that is absorbed by the other. The zero mass of photon explains the long-range nature of the Coulomb forces. According to Yukawa, because nuclear forces being of short-range kind, the quantum exchanged between the nucleons must have a finite mass. The mass of the quantum can be deduced from the range of nuclear forces and was calculated by Yukawa as ~300 m_e, where m_e is the mass of electron. The quanta involved in such interactions have been given name as p-mesons or pions. It was proposed that there exist three pions and they were denoted as positively charged pion p^+, neutral pion p^0 and negatively charged pion p^-. All these three pions were discovered and their properties have been measured.

To explain neutron–proton interactions, a neutron decays by emission proton and p^-, i.e.

$n \rightarrow p + p^-$ and this p^- is absorbed by another proton or $p^- + p \rightarrow n$. Similarly, proton–neutron interaction is explained by the emission of p^+, i.e. $p \rightarrow n + p^+$ and $n + p^+ \rightarrow p$.

For p–p interaction, $p \rightarrow p' + p^0$, where p' is the new proton by emitting p^0 and this p^0 is absorbed by another proton as $p^0 + p \rightarrow p'$ where p' is the new proton formed by absorbing p^0.

Similar equations can be written for n–n interaction.

In the case of single nucleon, the pions are constantly emitted and absorbed by the same nucleon, i.e. a nucleon is surrounded by a cloud of pions.

There is another question we may ask. When a pion is emitted, will there be an apparent non-conservation of energy by an amount $\Delta E = mp\ c^2$? The answer lies in the uncertainty principle, which allows non-conservation of energy by an amount given by ΔE

$$\Delta E\ \Delta t \sim \hbar$$

or

$$\Delta t \sim \frac{\hbar}{\Delta E} \approx \frac{\hbar}{m_\pi c^2} \leq 4 \times 10^{-24}\ \text{s} \tag{1.38}$$

If pion is travelling with approximately equal to that of light or c, then in time Δt, it will cover a distance

$$\Delta t \times c = \frac{\hbar}{m_\pi c} \approx 1.2 \times 10^{-13}\ \text{cm} \tag{1.39}$$

which is precisely the range of nuclear forces.

1.11 MASS SPECTROMETRY

Mass spectrometry also known as *mass spectroscopy* is an analytical technique used to measure the mass-to-charge ratio of ions. It is most generally used to find the composition of a physical sample by generating a mass spectrum representing the masses of sample components. The mass spectrum is obtained by a *mass spectrometer*.

Although the general term mass spectroscope is often used, a distinction can be made between *spectrometer* in which a metre measures the ion current, and a *spectrograph* in which the record appears on a photographic plate. The term mass spectrometer, however, is often used for all such instruments. A typical mass spectrometer consists of three parts, as shown in Figure 1.8, an ion source, a mass analyzer, and a detector and data analysis system. The stages within the mass spectrometer are:

1. Producing ions from the sample.
2. Separating ions of differing masses.
3. Detecting the number of ions of each mass produced.
4. Collecting the data and generating the mass spectrum.

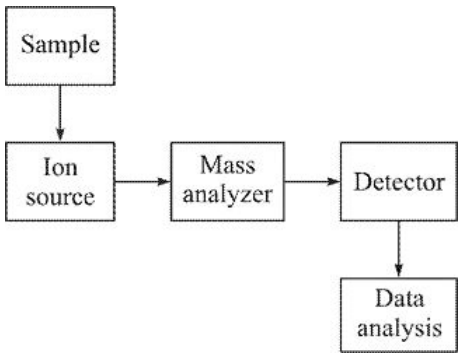


Figure 1.8 Block diagram of the mass spectrometer.

The foundations of mass spectrometry lie in the work of Thomson and Aston at the Cavendish Laboratories, Cambridge University. From 1897, the work was carried out by Thomson and his co-workers. Thomson's original work on the existence and properties of canal rays (positive ions) was taken up by Aston who demonstrated the existence of several isotopes of non-radioactive elements. Aston used electrostatic and magnetic fields to separate isotopes and focus them onto a photographic plate. Over the next few years a number of workers took up the development of mass spectrometry, including Dempster, Herzog, Bainbridge and Nier. By the end of the 1930s, mass spectrometry had become an established technique for two purposes:

(i) Determination of number of isotopes in a given element.
(ii) Determination of masses of various isotopes.

In the present discussion, we focus on two types of mass spectrographs:

(a) Bainbridge spectrograph.
(b) Bainbridge and Jordan mass spectrograph.

1.11.1 Bainbridge Spectrograph

In 1933, Bainbridge modified the existing mass spectrographs (like Aston's mass spectrograph, Dempster's mass spectrometer, etc.) and developed a new spectrograph known as Bainbridge spectrograph. It is based on the principle of velocity selecting/filtering. It contains an ion source, in which an anode *D* is fitted as shown in Figure 1.9. Cathode is perforated. The gas whose isotopes are to be investigated is admitted into the ion source through a gas tube marked as gas in the figure. A high potential difference (5 to 20 kV) is applied between the anode and the cathode (not shown in

figure). The high voltage between the anode and the cathode ionizes the gas present in the ion source. Ionization is further improved by bombarding the gaseous isotopes with a beam of electrons. Positive ions are accelerated towards the cathode. Because of the perforation in the cathode the accelerated ions come out and are allowed to pass through a slit S_1. Then they are made to travel through two plates A and B, between which an electric field (E) is applied. Here a magnetic field (strength B_1) is also applied at right angle to the electrostatic field. These two are adjusted in such a way that electrostatic and electromagnetic forces act in opposite direction to each other. Finally, positive ions pass through the second slit S_2. The force on the ions due to electric field is Ee and due to magnetic field B_1ev, where e is the charge present on the ions and v is the velocity of the ions. As these two forces are equal and opposite, therefore,

$$B_1ev = Ee \tag{1.40}$$

or

$$v = \frac{E}{B_1} \tag{1.41}$$

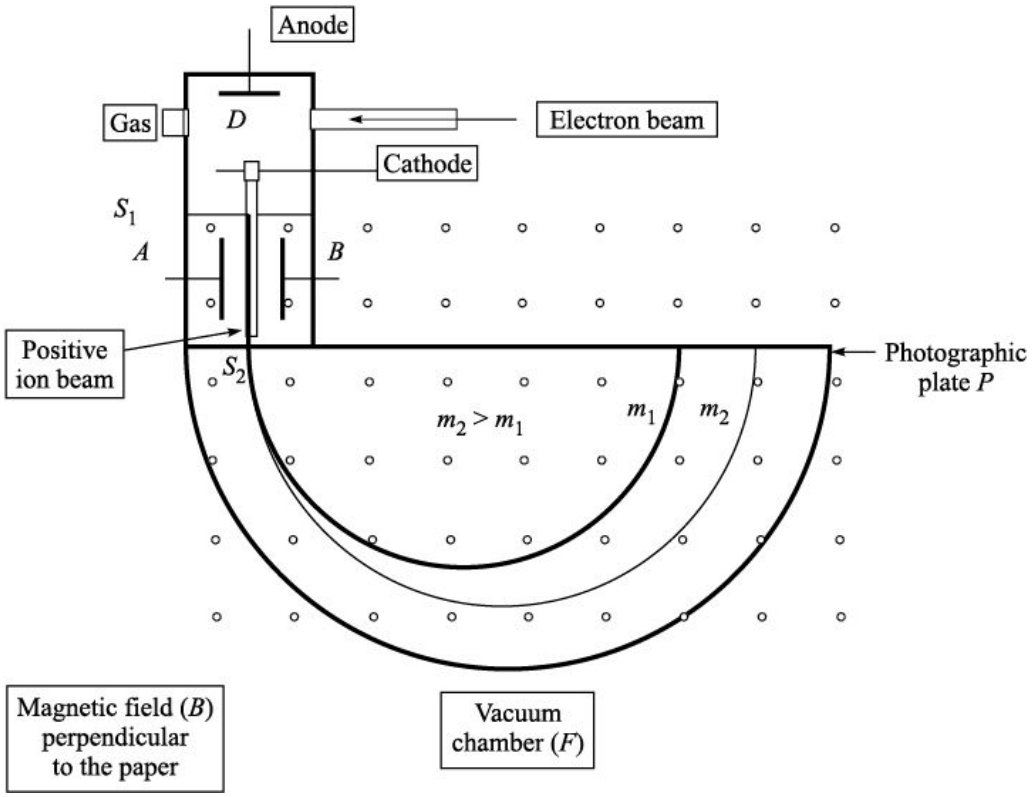


Figure 1.9 Schematic diagram of Bainbridge spectrograph.

Only those positive ions, whose velocity is given by Eq. (1.41), continue to move forward and pass through the slit S_2. Therefore, in this arrangement of cross magnetic and electric fields, positive ions having exactly the same velocity v are selected. For this reason, the combination of slits S_1 and S_2 and deflecting plates A and B is called velocity selector/filter. The ions moving with velocity v enter a vacuum chamber F, which is also called analyzing or deflecting chamber, where a constant magnetic field B is maintained at right angle to the direction of motion of ions. Under the influence of this magnetic field, the ions move in a semicircular path of radius r, given by

$$\frac{Mv^2}{r} = Bev$$

or

$$\frac{e}{M} = \frac{v}{rB}$$

Substituting v from Eq (1.41), we get

$$\frac{e}{M} = \frac{E}{BB_1 r} \tag{1.42}$$

Equation (1.42) shows that ions having different $\frac{e}{M}$ values, describe semicircular paths of different radii. Ultimately these ions strike photographic plate *P*. Equation (1.42) can also be written as

$$r = \frac{E}{BB_1}\frac{M}{e} = KM$$

where $K = E/BB_1e$ is constant, therefore, the radius of curvature of the path is directly proportional to the mass of the positive ions, provided the charge is same for all the ions. Suppose a given element contains two isotopes with masses m_1 and m_2, with $m_2 > m_1$. Atoms of the isotope with mass m_1 will describe a semicircle of smaller radius, while the atoms of the isotope with mass m_2 will move in a semicircle of larger radius. Thus, for a given element, particles having the same mass will follow a particular path and strike the photographic plate *P* at the same point.

In the first experiment, Bainbridge analyzed germanium vapours. He used a strong magnetic field (~15 kG) across the analyzing chamber and the semicircular path of the ions was about
40 cm. With this arrangement, he was able to identify five isotopes of germanium with mass numbers 70, 72, 73, 74 and 76.

Some of the advantages of Bainbridge mass spectrograph are as under:

- Since $r = KM$, so the mass scale is linear.
- The ion beam strikes the photographic plate normally, the error due to irregularities in the plate is negligible.
- It has a high precision.

1.11.2 Bainbridge and Jordan Mass Spectrograph

In 1936, Bainbridge and Jordan developed a mass spectrograph capable of mass determination with an accuracy of 1 in 30,000 parts. The earlier spectrographs were designed either to focus ions having the same velocity or moving in same direction to one point. However, in this mass spectrograph, all the ions having same e/M value are focused at the same point irrespective of the direction or velocity.

A schematic diagram of this mass spectrograph is shown in Figure 1.10. The ions from the discharge tube having anode A and a perforated cathode C are collimated by slits S_1 and S_2. They are then deflected through $p/\sqrt{2}$ radians (~127°), using the radial magnetic field to produce an energy spectrum in front of the third slit S_3. Beams of ions of all e/M values of this energy spectrum diverging from the focal plane of the electrical deflector are further deflected through a mean angle of $p/3$ radians by magnetic field, which forms a mass spectrum of focused ion beams on the photographic plate.

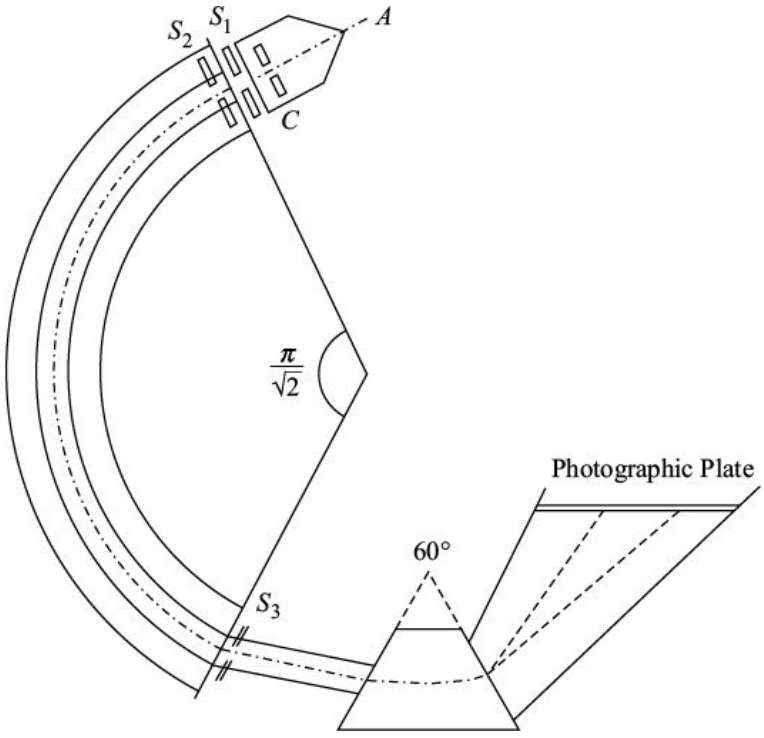


Figure 1.10 Schematic diagram of Bainbridge and Jordan mass spectrograph.

The focusing of ions of different velocities is obtained as a result of the geometrical arrangement of the electric and magnetic deflecting fields which are so placed that the dispersion of the ions produced by one field is cancelled by the dispersion produced by the other field for a given velocity difference. Under these conditions ions of the same e/M values having all velocities within a certain range are focused on the recording plate.

1.12 DETERMINATION OF CHARGE BY MOSELEY LAW

While working in Rutherford's laboratory at University of Manchester, in 1913, H. Moseley demonstrated that charge present on a nucleus could be determined by analyzing its X-ray spectrum. At that time, it was discovered that crystal could act as diffraction grating and hence crystals could be used to compare the wavelengths of X-rays. Moseley studied the characteristic X-rays of a number of elements. The 10-consecutive elements chosen by Moseley were $_{20}$Ca, $_{22}$Ti, $_{23}$V, $_{24}$Cr, $_{25}$Mn, $_{26}$Fe, $_{27}$Co, $_{28}$Ni, $_{29}$Cu and $_{30}$Zn. He obtained the photographs of X-rays lines. From the position of the lines on photographs, Moseley determined the frequencies of the corresponding X-rays. From this data, he concluded that there was one fundamental quantity Q which increases in regular steps as one passes from one element to the next, using the chemical order of the elements in the periodic table. He found that square root of the frequency of most intense X-ray (known as *Ka* line) is directly proportional to a quantity which Moseley called Q. Later on the quantity Q was called Z atomic number of the element. Moseley formulated a simple relation between Z and square root of the X-ray frequency as

$$\sqrt{f} = k(Z - k_2) \tag{1.43}$$

where f is the frequency of the most intense X-ray line, k and k_2 are constants. This formula is known as Moseley's law.
Equation (1.43) can also be written as

$$f = k_1(Z - k_2)^2 \tag{1.44}$$

where k_1 is another constant.

The constants k_1 and k_2 depend on the type of X-ray line. For *Ka* X-ray line (*Ka* is X-ray emitted when electron makes a transition from *L* shell to *K* shell), Moseley found $k_1 = 2.47 \times 10^{15}$ Hz, and $k_2 \approx 1$. Later on it was found that k_1 is actually product of Rydberg constant (3.29×10^{15}) and 3/4. The constants k_1 and k_2 are same for all *Ka* X-ray lines irrespective of which element is emitting the X-ray. Substituting these constants, Eq. (1.44) becomes

$$f_K a = 2.47 \times 10^{15}(Z - 1)^2 \text{ Hz.} \tag{1.45}$$

Similarly, for *La* X-ray line (*La* X-ray is emitted when an electron makes a transition from *M* shell to *L* shell), constants k_1 and k_2 were found to be $k_1 = 0.457 \times 10^{15}$ Hz, and $k_2 \approx 7.4$. Here constant k_1 is the product of Rydberg constant and 5/36. Therefore, for *La* X-ray line Moseley law becomes

$$f_L a = 0.457 \times 10^{15}(Z - 7.4)^2 \text{ Hz} \tag{1.46}$$

In terms of energy (as with modern X-ray spectrometers it is easy to measure energy than to calculate frequency of X-rays), Eqs. (1.45) and (1.46) become

$$E_K a(\text{keV}) = 1.042 \times 10^{-2}(Z - 1)^2$$

Similarly, for *La* line (which is generally less intense than *Ka*) the above relation is

$$E_L a(\text{keV}) = 1.494 \times 10^{-3}(Z - 7.4)^2$$

Thus, by experimentally measuring the energy (*E*) or frequency (*f*), *Z*—the atomic number of the element can be calculated. This method was first applied to newly discovered element hafnium (Hf), which is chemically identical to zirconium (Zr). C. Coster and G. von Hevesy in 1923 found the atomic number of hafnium as 72 from its X-ray spectrum.

NUMERICAL PROBLEMS

Section 1.3

Solved Problems

1. Calculate the momentum of a neutron having de Broglie wavelength of 5 fm.

Solution:

$$\text{Momentum } p = \frac{h}{\lambda}$$

$$\hbar = 1.055 \times 10^{-21} \text{ MeV s}$$

$$\lambda = 5 \text{ fm} = 5 \times 10^{-15} \text{ m}$$

Therefore,

$$p = \frac{1.055 \times 10^{-21}}{5 \times 10^{-15}}$$

$$= 0.211 \times 10^{-6} \text{ MeV s/m}$$

2. Group the following nuclides as isotopes, isotones and isobars:

$$^{12}_{6}\text{C},\ ^{13}_{6}\text{C},\ ^{14}_{6}\text{C},\ ^{14}_{7}\text{N},\ ^{14}_{8}\text{O},\ ^{15}_{7}\text{N},\ ^{15}_{8}\text{O},\ ^{16}_{6}\text{C},\ ^{16}_{7}\text{N},\ ^{16}_{8}\text{O},\ ^{17}_{7}\text{N},\ ^{17}_{8}\text{O}$$

Solution:

Isotopes:

$^{12}_{6}\text{C},\ ^{13}_{6}\text{C},\ ^{14}_{6}\text{C},\ ^{16}_{6}\text{C}$

$^{14}_{7}\text{N},\ ^{15}_{7}\text{N},\ ^{16}_{7}\text{N},\ ^{17}_{7}\text{N}$

$^{14}_{8}\text{O},\ ^{15}_{8}\text{O},\ ^{16}_{8}\text{O},\ ^{17}_{8}\text{O}$

Isotones:

$^{12}_{6}\text{C},\ ^{14}_{8}\text{O},$

$^{13}_{6}\text{C},\ ^{14}_{7}\text{N},\ ^{15}_{8}\text{O}$

$^{14}_{6}\text{C},\ ^{15}_{7}\text{N},\ ^{16}_{8}\text{O}$

$^{16}_{7}\text{N},\ ^{17}_{8}\text{O}$

Isobars:

$^{14}_{6}\text{C},\ ^{14}_{7}\text{N},\ ^{14}_{8}\text{O}$

$^{15}_{7}\text{N},\ ^{15}_{8}\text{O}$

$^{16}_{6}\text{C},\ ^{16}_{7}\text{N},\ ^{16}_{8}\text{O}$

$^{17}_{7}\text{N},\ ^{17}_{8}\text{O}$

Unsolved Problems

1. Calculate de Broglie wavelength for a proton having energy 150 MeV. [**Ans.** 0.596 fm]
2. Calculate the ratio of de Broglie wavelengths of a proton and electron having energy equal to 150 MeV. [**Ans.** 0.462]
3. Calculate the ratio of momenta of a proton and neutron having the same de Broglie wavelength. [**Ans.** 0.999]
4. Find the number of neutrons present in $^{238}_{92}$U nucleus. [**Ans.** 146]
5. Find the number of electrons present on $^{238}_{92}$U atom. [**Ans.** 92]
6. Thermal neutrons have an average kinetic energy $\frac{3}{2}kT$, where k is Boltzmann's constant and T is room temperature, 300 K. (i) What is the average energy in eV of a thermal neutrons? (ii) What is the corresponding de Broglie wavelength? [**Ans.** (i) 0.038 eV, (ii) 1.46 □ 10^{-10} m]

Section 1.4

Solved Problems

1. Calculate the energy of electron at rest.

Solution:

$$\begin{aligned}\text{Mass of the electron } m &= 9.1 \times 10^{-31} \text{ kg}\\ \text{Energy } mc^2 &= 9.1 \times 10^{-31} \times (3 \times 10^8)^2 \text{ J}\\ &= 8.19 \times 10^{-14} \text{ J}\\ 1.6 \times 10^{-13} \text{ J} &= 1 \text{ MeV}\\ 81.9 \times 10^{-15} \text{ J} &= \frac{8.19 \times 10^{-14}}{1.6 \times 10^{-13}} \text{ MeV}\\ &= 0.511 \text{ MeV}\end{aligned}$$

2. Estimate the A value and identify the nucleus if its radius is given to be 3.46 fm.

Solution: Radius of the nucleus $R = 3.46$ fm
We have from Eq. (1.3)

$$r = r_0 A^{1/3}$$

and

$$r_0 = 1.2 \text{ fm}$$

Therefore,

$$r = r_0 A^{1/3}$$

and

$$r_0 = 1.2 \text{ fm}$$

Therefore,

$$\frac{r}{r_0} = A^{1/3}$$

$$A^{1/3} = \frac{r}{r_0}$$

$$= \frac{3.46}{1.2}$$

$$= 2.883$$

$$A = (2.883)^3$$

$$= 23.97$$

$$\approx 24$$

Therefore, $A = 24$, the nucleus is ^{24}Mg.

3. Assume 1 amu = 1.66 □ 10^{-27} kg, estimate the density of nuclear matter.

Solution: Protons and neutrons present in the nucleus constitute nuclear matter. Let mass of a nucleus $A = 40$ amu.

$$\text{Mass of this nucleus } m = 40 \times 1.66 \times 10^{-27} \text{ kg}$$
$$= 6.64 \times 10^{-26} \text{ kg}$$
$$\text{Radius } r = r_0 A^{1/3}, \text{ where } r_0 = 1.2 \text{ fm and } A = 40$$
$$\therefore \quad r = 1.2 \times 40^{1/3} \times 10^{-15} \text{ m}$$
$$= 4.1 \times 10^{-15} \text{ m}$$
$$\text{Volume of nucleon } V = \frac{4}{3}\pi r^3$$
$$= \frac{4}{3} \times 3.1415926 \times (4.1 \times 10^{-15})^3 \text{ m}^3$$
$$= 2.886 \times 10^{-43} \text{ m}^3$$
$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{m}{V}$$
$$= \frac{6.64 \times 10^{-26}}{2.886 \times 10^{-43}}$$
$$= 2.3 \times 10^{17} \text{ kg/m}^3$$
$$\therefore \quad \text{Density of nuclear matter} = 2.3 \times 10^{17} \text{ kg/m}^3$$

4. Find the nuclear density of ^{235}U, if $r_0 = 1.2$ fm.

Solution:

$$\text{Mass of 1 nucleon} = 1.66 \times 10^{-27} \text{ kg}$$
$$\text{Mass of 235 nucleons } m = 235 \times 1.66 \times 10^{-27} \text{ kg}$$
$$= 3.901 \times 10^{-25} \text{ kg}$$
$$\text{Radius of } ^{235}\text{U nucleus } r = r_0 A^{1/3}$$
$$= 1.2 \times 10^{-15} \times 235^{1/3}$$
$$= 7.405 \times 10^{-15} \text{ m}$$

$$\text{Volume of } {}^{235}\text{U nucleus } V = \frac{4}{3}\pi r^3$$

$$= \frac{4}{3} \times 3.1415926 \times (7.405 \times 10^{-15})^3 \text{ m}^3$$

$$= 1.701 \times 10^{-42} \text{ m}^3$$

$$\text{Density of } {}^{235}\text{U} = \frac{\text{Mass}}{\text{Volume}}$$

$$= \frac{3.901 \times 10^{-25}}{1.701 \times 10^{-42}} \text{ kg/m}^3$$

$$= 2.29 \times 10^{17} \text{ kg/m}^3$$

5. The radii of oxygen and lead nuclei are found to be 3 fm and 7 fm respectively. Their masses are 2.7 □ 10^{-26} kg and 3.4 □ 10^{-25} kg respectively. Calculate their densities.

Solution:

$$\text{Volume of nucleus with mass number } A = \frac{4}{3}\pi r^3$$

$$\text{For oxygen nucleus volume} = \frac{4}{3} \times 3.14159 \times (3 \times 10^{-15})^3$$

$$= 1.131 \times 10^{-43} \text{ m}^3$$

$$\text{Density of oxygen nucleus} = \frac{\text{Mass}}{\text{Volume}}$$

$$= \frac{2.7 \times 10^{-26}}{1.131 \times 10^{-43}}$$

$$= 2.38 \times 10^{17} \text{ kg/m}^3$$

$$\text{For lead nucleus volume} = \frac{4}{3} \times 3.14159 \times (7 \times 10^{-15})^3$$

$$= 1.437 \times 10^{-42} \text{ m}^3$$

$$\text{Density of lead nucleus} = \frac{\text{Mass}}{\text{Volume}}$$

$$= \frac{3.4 \times 10^{-25}}{1.437 \times 10^{-42}}$$

$$= 2.37 \times 10^{17} \text{ kg/m}^3$$

6. If the energy of the a-particle emitted by ^{241}Am is 5.48 MeV, find the closest distance it can approach a Au nucleus.

Solution: The distance of closest approach is given by the relation

$$d = \frac{2Ze^2}{4pe_0E}$$

Given:

$$E = 5.48 \text{ MeV}$$

$$= 5.48 \square 1.6 \square 10^{-13} \text{ J}$$

Therefore,

$$d = \frac{8.98 \times 10^9 \times 2 \times 79 \times (1.6 \times 10^{-19})^2}{5.48 \times 1.6 \times 10^{-13}}$$

$$= 4.14 \square 10^{-14} \text{ m}$$

7. Determine the radius of ^{208}Pb.

Solution: For ^{208}Pb, $A = 208$

We have

$$r = r_0 A^{1/3} \qquad \text{and} \qquad r_0 = 1.2 \square 10^{-15} \text{ m}$$

Therefore,

$$\text{Radius } r = 1.2 \square 10^{-15} \square 208^{1/3} = 7.11 \square 10^{-15} \text{ m}$$

or

$$\text{Radius of } ^{208}\text{Pb} = 7.11 \square 10^{-15} \text{ m} = 7.11 \text{ fm}$$

Unsolved Problems

1. It is possible on the basis of shell model that nucleus with $Z = 110$ and $A = 294$ may be exceptionally long-lived. Estimate its nuclear radius. [**Ans.** 7.98 fm]

2. Ordinary boron is a mixture of $^{10}_{5}$B and $^{11}_{5}$B isotopes and has composite atomic mass of 10.82 amu. What percentage of each isotope is present in ordinary boron? Given $m_{B10} = 10.012937$ amu and $m_{B11} = 11.009305$ amu. [**Ans.** 18.99, 81.01]

3. Find the distance of closest approach, when

(i) 5 MeV protons are bombarded on $^{27}_{13}Al$.

(ii) 5 MeV deuterons are bombarded on $^{27}_{13}Al$. [**Ans.** (i) 3.5 □ 10^{-15} m, (ii) 3.5 □ 10^{-15} m]

4. Find the distance of closest approach, when

(i) 50 MeV protons are bombarded on $^{208}_{82}Pb$.

(ii) 50 MeV are bombarded on $^{208}_{82}Pb$. [**Ans.** (i) 2.87 □ 10^{-17} m, (ii) 1.9 □ 10^{-13} m]

5. What is the energy equivalent to the mass of an electron and a proton? [**Ans.** 0.511 MeV, 939.3 MeV]

6. The density of nuclear matter is 2.294 □ 10^{17} kg/m^3. Estimate the energy equivalent to 1Å^3 of nuclear matter. [**Ans.** 20.646 kJ]

Section 1.5

Solved Problems

1. Calculate the binding energy of *a*-particle and express it in MeV and joules. Given
m_p = 1.00758 amu, m_n = 1.00897 amu and m_{He} = 4.0028 amu.

Solution: Binding energy of *a*-particle is given by Eq. (1.18)

$$BE = (Zm_p + Nm_n - m_{He})\text{ amu}$$
$$= (2 \square 1.00758 + 2 \square 1.00897 - 4.0028)\text{ amu}$$
$$= 0.0293\text{ amu}$$

Now,

$$1\text{ amu} = 931.49\text{ MeV}$$
$$0.0293\text{ amu} = 0.0293 \square 931\text{ MeV}$$
$$= 27.29\text{ MeV}$$

Therefore, the binding energy of *a*-particle = 27.29 MeV

$$1\text{ amu} = 1.49239 \square 10^{-10}\text{ J}$$
$$0.0293\text{ amu} = 0.0293 \square 1.49239 \square 10^{-10}$$
$$= 4.37 \square 10^{-12}\text{ J}$$

2. Find the energy required in joules to break ^{12}C into 3 *a*-particles. The atomic mass of ^{12}C = 12 amu and m_{He} = 4.0026 amu.

Solution: We have the equation

$$^{12}_{6}C \rightarrow 3\,^{4}_{2}He$$

Given:

Mass of ^{12}C = 12 amu

Mass of *a*-particle = 4.0026 amu

Mass of 3*a*-particle = 3 × 4.0026 amu = 12.0078 amu

Difference in two masses = 12 – 12.0078 = –0.0078 amu

1 amu = 1.49239×10^{-10} J

0.0078 amu = $-0.0078 \times 1.49239 \times 10^{-10} = -1.16 \times 10^{-12}$ J

Therefore, 1.16×10^{-12} J of energy is required to break up ^{12}C into 3 *a*-particles.

3. Masses of He nucleus, proton and neutron in amu are 4.0026, 1.007895 and 1.008665. Find the energy required to knock out nucleons from the He nucleus.

Solution: We have the equation

$$^{4}_{2}He \rightarrow 2p + 2n$$

Given:

m_{He} = 4.0026 amu

m_p = 1.007895 amu

m_n = 1.008665 amu

Mass of 2 protons and 2 neutrons = 2 × 1.007895 + 2 × 1.008665 = 4.03312 amu

Therefore, mass defect = 4.0026 – 4.03312 = –0.03052 amu

Equivalent energy = –0.03052 × 931.49 = –28.43 MeV

Energy required to knock out 2 protons and 2 neutrons from He nucleus = 28.43 MeV.

4. Find the binding energy of $^{56}_{26}Fe$ in MeV. Given: m_p = 1.007825 amu, m_n = 1.008665 amu and m_{Fe} = 55.934939 amu.

Solution: Binding energy (BE) is given as

$$BE = (26m_p + 30m_n - m_{Fe}) \text{ amu}$$

$$= (26 \square 1.007825 + 30 \square 1.008665 - 55.934939) \text{ amu}$$

or

$$\text{Binding energy } BE = 0.5285 \text{ amu}$$

or

$$\text{Binding energy } BE = 0.5285 \square 931.49 = 492.29 \text{ MeV}$$

5. The mass of a deuteron is 2.014103 amu. If the masses of proton and neutron are respectively 1.007825 amu and 1.008663 amu, find the mass defect and packing fraction.

Solution:

$$\text{Mass defect } \Delta M = m_p + m_n - m_d$$

$$= 1.008825 + 1.008663 - 2.014104 = 0.002385 \text{ amu}$$

$$= 2.22 \text{ MeV}$$

$$\text{Packing fraction } f = \frac{\text{Mass} - A}{A}$$

$$= \frac{2.014103 - 2}{2}$$

$$= 0.00705$$

6. Calculate the average binding energy per nucleon of ^4He nucleus. Given m_{He} = 4.002643 amu, m_p = 1.007825 amu and m_n = 1.008665 amu.

Solution: Binding energy is given by

$$BE = (2m_p + 2m_n - m_{He}) \square 931.47 \text{ MeV}$$

$$= (2 \square 1.007825 + 2 \square 1.008665 - 4.002634) \square 931.47 \text{ MeV}$$

$$= 0.03034 \square 931.47 = 28.3 \text{ MeV}$$

$$\text{Binding energy per nucleon} = \frac{28.3}{4} = 7.07 \text{ MeV}$$

Unsolved Problems

1. Find the binding energy of $^{20}_{10}$Ne and of $^{79}_{35}$Br. Given m_p = 1.007825 amu, m_n = 1.008665 amu, m_{Ne} = 19.992436 amu and m_{Hr} = 78.918336 amu. [**Ans.** 160.64 MeV, 686.29 MeV]

2. Find the energies needed to remove a neutron from $^{4}_{2}$He, then to remove a proton, and finally to separate the remaining neutron and proton. Compare with the total binding energy of $^{4}_{2}$He, Given m_p = 1.007825 amu, m_n = 1.008665 amu, m_{He4} = 4.002603 amu, m_{He3} = 3.016029 amu and m_D = 2.014102 amu. [**Ans.** 20.58 MeV, 5,491 MeV, 2.224 MeV]

3. Which nucleus is more stable $^{7}_{3}$Li or $^{8}_{3}$Li? Given m_p = 1.007825 amu, m_n = 1.008665 amu, m_{Li7} = 7.016003 amu and m_{Li8} = 8.022486 amu. [**Ans.** $^{7}_{3}$Li is more stable as binding energy per nucleon for $^{7}_{3}$Li is 5.60 MeV/n and that of $^{8}_{3}$Li is 5.16 MeV/n]

4. The binding energy of $^{20}_{10}$Ne is 160.64 MeV. Find its atomic mass. Given m_p = 1.007825 amu and m_n = 1.008665 amu. [**Ans.** 19.99 amu]

5. Find the average binding energy per nucleon in $^{40}_{20}$Ca. Given m_p = 1.007825 amu, m_n = 1.008665 amu and m_{Ca} = 39.962591 amu. [**Ans.** 8.55 MeV/n]

6. By referring to Figure 1.4, determine whether energy must be supplied or will be released if we assume that the following transmutations can be accomplished: Fe □ Ag,
H □ He, He □ O and Pb □ Si. [**Ans.** supply, released, released, released]

7. Calculate the total binding energy of *d* and $^{20}_{10}$Ne and also calculate the binding energy per nucleon for these two nuclei. Given m_p = 1.007825 amu, m_n = 1.008665 amu m_d = 2.014102 amu and m_{Ne}= 19.992439 amu. [**Ans.** 2.2 MeV, 160.6 MeV, 1.1 MeV, 8.0 MeV]

8. What are the atomic and mass numbers of the oxygen isotope with 17 nucleons? Calculate the mass defect, binding energy and binding energy per nucleon for this nuclide, with the assumption that the mass defect is associated with the nucleus. Given $m_{^{17}O} = 16.999133$ amu, m_p = 1.007825 amu and m_n = 1.008665 amu. [**Ans.** 8, 17, 0.141452 amu, 131.76 MeV, 7.75 MeV/nucleons]

9. Natural oxygen contains three isotopes with atomic masses in amu of 15.9949, 16.9991 and 17.9992, and relative abundances of 2500**:**1**:**5. Determine to three decimal places the average atomic mass of oxygen. [**Ans.** 15.999 amu]

Section 1.6

Solved Problems

1. One nucleon is present in *p* state ($\square_1 = 1$) and another is in *d* state ($\square_2 = 2$). What will be the resulting orbital angular momenta, if both the nucleons coupled together?

Solution: $L = \square_1 + \square_2$

or *L* values will be between

$|\square_1 - \square_2|$ to $|\square_1 + \square_2|$

i.e. *L* will be 1, 2 or 3.

2. A proton is present in *f* state ($\square = 3$). What will be the total angular momentum of this proton?

Solution: $\left|\ell - \frac{1}{2}\right|$ to $\left|\ell + \frac{1}{2}\right|$

or $\frac{5}{2}$ and $\frac{7}{2}$

Unsolved Problems

1. Assume neutron has a radius of 1.5 fm and has uniform density. Calculate its frequency, if its angular momentum is $\hbar\sqrt{s(s+1)}$, where *s*

$= 1/2$, $\hbar = 1.05 \square 10^{-34}$ J s and

$m_n = 1.67 \square 10^{-27}$ kg. [**Ans.** $38.5 \square 10^{21}$ Hz]

2. Principle quantum number of a nucleus is 3. What are the possible values of the orbital quantum number $\square$? [**Ans.** 2, 1, 0]

Section 1.11

Solved Problems

1. A singly charged positive ion is accelerated through a potential difference of 1000 V in a mass spectrograph. It then passes through a uniform magnetic field B = 1500 gauss, and then deflected into a circular path of radius 0.122 m.

(i) What is the speed of the ion?
(ii) What is the mass of the ion?
(iii) What is the mass number of the ion?

Solution:

(i) Using the relations $\frac{mv^2}{R} = eVB$ and $\frac{1}{2} mv^2 = eV$, where e is ionic charge and V is the potential applied, we get

$$v = \frac{2V}{RB}$$

Now,

$$V = 1000 \text{ V}$$
$$R = 0.122 \text{ m}$$
$$B = 1500 \times 10^{-4} \text{ tesla}$$

Therefore,

$$v = \frac{2 \times 1000}{0.122 \times 1500 \times 10^{-4}} \text{ m}$$

or

$$= 1.093 \times 10^5 \text{ m/s}$$

(ii) Mass of the ion (in amu) $M = \frac{2eV}{v^2}$

$$= \frac{2 \times 1.602 \times 10^{-19} \times 1000}{(1.093 \times 10^5)^2}$$

$$= 2.682 \times 10^{-26} \text{ kg}$$

(iii) Mass number $= \frac{2.682 \times 10^{-26}}{1.673 \times 10^{-27}} \approx 16.03 \approx 16$

$= 16$

2. In a Bainbridge mass spectrograph, singly charged argon ions enter a velocity filter, where the electric field is 5 □ 10^4 V/m and magnetic field is 0.4 tesla. What is the velocity of the ions emerging the velocity filter? If the ions enter the magnetic field of 0.8 tesla, what will be the distances between the ions focused on the photographic plate for the three isotopes: ^{36}Ar, ^{38}Ar and ^{40}Ar?

Solution: Velocity of the ions emerging the velocity filter from Eq. (1.34) $v = \frac{E}{B_1}$ Here,

$$E = 5 \square 10^4 \text{ V/m}$$

$$B_1 = 0.4 \text{ tesla}$$

Therefore,

$$v = \frac{5 \times 10^4}{0.4} = 1.25 \times 10^5 \text{ m/s}$$

When the ions enter the magnetic field, the radius of the path is

$$R = \frac{Mv}{Be}$$

Here,

M = mass of the ion in kg

B = 0.8 tesla

For ^{36}Ar ions:

$$R = \frac{36 \times 1.673 \times 10^{-27} \times 1.25 \times 10^{5}}{0.8 \times 1.602 \times 10^{-19}}$$

$$= 0.0587 \text{ m} = 58.7 \text{ mm}$$

For ^{38}Ar ions:

$$R = \frac{38 \times 1.673 \times 10^{-27} \times 1.25 \times 10^{5}}{0.8 \times 1.602 \times 10^{-19}}$$

$$= 0.0620 \text{ m} = 62.0 \text{ mm}$$

For ^{40}Ar ions:

$$R = \frac{40 \times 1.673 \times 10^{-27} \times 1.25 \times 10^{5}}{0.8 \times 1.602 \times 10^{-19}}$$

$$= 0.0653 \text{ m} = 65.3 \text{ mm}$$

Therefore, distance between successive lines due to three different isotopes of Ar

$$= 2 \times (62.0 - 58.7) = 6.6 \text{ mm}$$

or

$$= 2 \times (65.3 - 62.0) = 6.6 \text{ mm}$$

Unsolved Problems

1. In a Bainbridge mass spectrograph, singly charged natural silver ions (^{107}Ag and ^{109}Ag) enter a velocity filter. In the velocity filter electric and magnetic fields are 5 □ 10^4 V/m and 0.4 tesla respectively. What is the speed of the ions emerging the velocity filter? If the ions after leaving velocity filter enter the magnetic field of 0.7 tesla, what is the separation between the two lines due to two isotopes of silver on a photographic plate? [**Ans.** 1.25 □ 10^5 m/s 7.6 mm]
2. Suppose that singly charged ions with masses close to 12 and 14 amu are accelerated through a potential difference of 1000 V and then travel through a magnetic field of
900 gauss. Where should collector plates for the two different ions be located? [**Ans.** 0.176 m, 0.190 m]

3. In a Bainbridge mass spectrometer, singly ionized ^{20}Ne passes into the deflection chamber with an initial velocity of 1.2 □ 10^5 m/s. Here it is deflected by a magnetic field of flux density 0.06 tesla. Calculate the radius of ionic path. Also calculate the radius of the path of singly ionized ^{22}Ne having the same initial velocity. [**Ans.** 0.418 m, 0.460 m]

REVIEW QUESTIONS

Short Answer Type

1. What is electric quadrupole moment?

2. What are thermal neutrons?

3. Explain the terms:
(i) Nuclear spin.
(ii) Nuclear magnetic dipole moment.
(iii) Nuclear electric quadrupole moment.

4. What are the main features of nuclear forces?

5. Discuss:
(i) Nuclear electric quadrupole moment.
(ii) Nuclear density.
(iii) Uncertainty principle forbids electrons in the nucleus.

6. Give any two evidences of neutron–proton model of the nucleus.

7. Does the nucleus of the atom has sharp boundaries? How is the mass distributed in the nucleus?

8. Show that nuclear density is same for all the nuclei.

9. Why the number of neutrons tends to exceed the number of protons in stable nuclei?

10. What are the reasons for assuming spin of electrons?

11. What is nuclear magneton m_n?

12. What is parity?

13. Why are even–even nuclei most stable?

14. Why does nuclear radius depend on mass number? Calculate its value for ^{27}Al.

15. List different types of magnetic moments that can be associated with nucleons.

16. Predict quadrupole moment for a nucleon whose ground state spin is /2. What is the significance of quadrupole moment?

17. Calculate nuclear density.

18. Explain the non-existence of electrons in the nucleus on the basis of magnetic moments.

19. Explain the concept of nuclear magnetic dipole moment.

20. Explain the concept of nuclear quadrupole moment.

21. What is average binding energy and discuss its variation with A?

22. What do you mean by even–even nuclei?

23. Obtain the ratio of densities of $^{197}_{79}\text{Au}$ and $^{107}_{47}\text{Ag}$.

24. What do you mean by charge independence of nuclear force?

25. Nuclear forces are of short-range. Explain.

26. What is an amu?

27. Explain the terms:
(i) Binding energy.
(ii) Mass defect.

28. Find a relation between electron volt and amu.

29. Why do we say that nuclear forces are of short-range?

30. Give one reason as to why electrons cannot exist in nucleus.

31. Why the density of nucleus is large?

32. Which is more, atomic binding energy or nuclear binding energy? Why?

33. What is the unit to measure the size of the nucleus?

34. What is packing fraction?

35. What type of statistics is followed by nucleons?

36. How does the binding energy curve explain fusion?

37. What are amu and electron volt?

38. What is nuclear magnetic moment?

39. What do you mean by odd–odd, odd–even and even–even nuclei? Draw the stability curve.

Long Answer Type

1. Name the forces existing between nucleons. Explain their characteristics.
2. Prove from statistics of angular momentum and wave mechanical considerations that electrons cannot exist in the nucleus.
3. Explain the terms: mass defect, binding energy and amu. Discuss how binding energy varies with A.
4. What is proton–neutron hypothesis? Give a reason for the acceptance of this hypothesis. Why do not electrons exist in the nucleus?
5. Explain the terms: mass defect and binding energy of the nucleus. Give the method of calculation of binding energy for a nucleus. Also give main features of binding energy per nucleon versus mass number graph.
6. Discuss magnetic moment of the nucleus. Explain how neutrons have magnetic moment.
7. Explain, giving two reasons, why the electron–proton hypothesis is not acceptable, while neutron–proton is.
8. What are the forces responsible for holding nucleons together in a nucleus? Give their main characteristics, explaining their origin.
9. Explain the non-existence of electrons inside the nucleus on the basis of wave mechanical considerations.
10. Enumerate the salient features of nuclear forces. How are they different from gravitational and electromagnetic forces?
11. Explain the relationship of nuclear mass with nuclear size. Show how binding energy changes with mass number.
12. Give the reasons for non-existence of electrons in the nucleus.
13. What is binding energy? Explain the stability of the nucleus on the basis of this concept.
14. Define binding energy and explain what one can learn from the variation of binding energy with A.
15. Write short notes on:
    (i) Nuclear magnetic moment.
    (ii) Nuclear quadrupole moment.
16. Give one method to determine the size of the nucleus.

17. Calculate nuclear mass density and discuss its variation with distance from the centre of the nucleus.

18. Explain meson or Yukawa theory of nuclear forces.

19. Write notes on:
(i) Proton–neutron theory.
(ii) Fermi–Dirac statistics.
(iii) Nuclear magnetic dipole moment.

20. Discuss four reasons for the failure of proton–electron hypothesis of a nucleus.

21. What are the reasons which lead to the belief that spin is associated with electrons?

22. State the main properties of nuclear forces. Explain the hypothesis of charge and spin independence of nuclear forces.

23. Explain nuclear binding energy and packing fraction. Discuss graphically the variation of average binding energy per nucleon with A.

24. Assuming nucleus to be uniformly charged, calculate the work done against electrostatic forces in giving it a charge Ze.

25. Discuss the meson theory of nuclear forces. Give the main properties of these forces.

26. What is electric quadrupole moment of a nucleus? Discuss the shapes of the nucleus on its basis.

Chapter 2

Nuclear Models

2.1 INTRODUCTION

As discussed in Chapter 1, the size of the nucleus is very small and nuclear forces are far more complicated than other well-known forces. In fact, the picture of nuclear forces is still not clear. This picture is different from the case of atom, where the forces are known and atomic model is well established for deducing various properties in atomic domain. Due to the lack of detailed knowledge of nuclear forces, nuclear models, namely liquid drop model, shell model, Nilson model, Fermi gas model, collective model, Bohr Motelson model, interacting boson model, etc. have been developed, each of which is useful in a more or less limited fashion.

In order to understand and predict the properties of the nucleus, we have to know the forces completely. For knowing nuclear forces, we adopt a different approach. In nuclei, we choose an oversimplified theory, the treatment of which is mathematically possible, but the theory should be rich in physics. If this theory is fairly successful in accounting for at least a few properties of the nucleus, we can then improve the model by adding additional terms so that it is capable to account more nuclear properties. In this way, we construct a nuclear model, a simplified view of nuclear structure, which still contains the essentials of the nuclear properties. A good nuclear model must satisfy following two criteria:

- It must reasonably well account for previously measured nuclear properties.
- It must predict additional nuclear properties that can be measured in new experiments.

The development of nuclear models has taken place along the following lines.

In the first type of nuclear models, nucleus has been treated like a drop of liquid, in which nucleons present in the nucleus interact very strongly among themselves. This is like molecules present in a drop of liquid, which interact among themselves very strongly. This treatment gave rise to models like *liquid drop model*, *collective model,* etc. The second type of models is constructed in analogy with the shell model of the atom. In these models, the nucleons are weakly interacting among themselves. This treatment gave rise to *Fermi gas model, shell model, Nilson model,* etc.

In this chapter, we discuss only two models, i.e. liquid drop model and shell model. In the end, a brief description of Fermi gas model is given.

2.2 LIQUID DROP MODEL

Weizsacker in 1935 proposed on the basis of experimental facts that a nucleus resembles a drop of liquid. In 1939, Bohr and Wheeler further developed this model to explain the phenomenon of nuclear fission.

Following are some of the similarities between a drop of liquid and nucleus, which prompted Weizsacker to develop the liquid drop model.

Similarities between Liquid Drop and Nucleus

1. Nuclear forces are analogous to the surface tension of a liquid.
2. The nucleons behave in a manner similar to that of molecules in a liquid drop.
3. The density of the nuclear matter is almost independent of A, showing resemblance to liquid drop where the density of a liquid is independent of the size of the drop.
4. The constant binding energy per nucleon is analogous to the latent heat of vaporization.
5. The disintegration of nuclei by the emission of particles is analogous to the evaporation of molecules from the surface of liquid.
6. The absorption of bombarding particles by a nucleus corresponds to the condensation of drops.
7. The energy of nuclei corresponds to internal thermal vibrations of drop molecules.

Based on these similarities, Weizsacker in 1935 and Bohr and Wheeler in 1939 developed liquid drop model. They ignored the finer features of nuclear forces but strong internucleon attraction is stressed.

Assumptions of the Liquid Drop Model

1. The nucleus consists of incompressible matter.
2. The nuclear force is identical for every nucleon.
3. The nuclear force saturates.
4. In an equilibrium state, the nuclei of atom remain spherically symmetric under the action of strong attractive nuclear forces.

2.2.1 Semiempirical Mass Formula

The analogy between nucleus and liquid drop has been used to set up a semiempirical formula for mass (or binding energy) of a nucleus in its ground state. The formula has been obtained by considering different factors of the nucleus binding.

The mass of the nucleus can be expressed in terms of the total binding energy B and the masses of Z protons and N neutrons as

$$\boxed{M = ZM_p + NM_n - B} \tag{2.1}$$

The binding energy B of a nucleus is given by the sum of five terms as

$$B = B_1 + B_2 + B_3 + B_4 + B_5 \tag{2.2}$$

which are explained in the following sections.

Volume Energy Term (B_1)

The volume term arises from the interaction of the nucleons through the strong force. When a liquid drop evaporates, the energy required for this process is the product of mass of the drop M_m and latent heat of vaporization L. This energy is used to break all the molecular bonds. This is same as the binding energy of the drop B. So

$$B = LM_m N \tag{2.3}$$

where N is the number of molecules in the drop. Equation (2.3) can also be written as

$$\frac{B}{N} = LM_m = \text{constant} \tag{2.4}$$

This means that B/N is independent of the number of molecules present in the liquid drop. As we know that in the liquid drop, a molecule interacts only with its nearest neighbours and number of neighbours is independent of the size of the drop. This characteristics of the system shows that range of interaction among the molecules is much smaller than the dimensions of the drop.

In Chapter 1, we have seen that neutrons and protons are held together in nuclei by short-range attractive forces. These forces reduce the mass of the nucleus below that of its constituents by an amount proportional to the number of nucleons A. Since the volume of the nucleus is proportional to A, hence this term is regarded as a volume binding energy and in analogy to Eq. (2.4) is given by

$$\boxed{B_1 = a_v A} \tag{2.5}$$

where a_v is a proportionality constant and subscript v is for volume.

Surface Energy Term (B_2)

The surface term is a correction to the volume term to take into account that the nucleons at the surface of the nucleus do not have the same level of interactions as nucleons in the interior of the nucleus. In the above discussion, we have assumed that all the molecules are surrounded by its neighbours, while in actual practice the molecules at the surface do not have any neighbours on all the sides. So these molecules are not as tightly bound as the molecules in the interior. Extending this argument to the nuclear case, some nucleons are nearer to the surface, and so they interact with fewer nucleons. Thus, the binding energy is reduced by an amount proportional to the surface area of the nucleus of radius r as the nucleons on the surface are less tightly bound than those in the interior. This term is proportional to the surface area of the nucleus of radius $r(= r_0 A^{1/3})$. Therefore,

$$B_2 \propto -4\pi r^2$$

or

$$B_2 \propto -4p\, r^2{}_0\, A^{2/3}$$

which is usually expressed as

$$\boxed{B_2 = -a_s A^{2/3}} \tag{2.6}$$

where negative sign is for decrease in energy and a_S is constant.

Coulomb Energy Term (B_3)

The Coulomb term represents the energy incorporated in the nucleus as a result of the positive charge present in the nucleus. The only long-range force in the nucleus is the Coulomb repulsion between protons. The total work done in assembling a nucleus consisting of Z protons is given by

$$W = \frac{\frac{3}{5} Z^2 e^2}{4\pi\varepsilon_0 r}$$

where r is the radius of the nucleus.

For a single-proton nucleus

$$w = \frac{\frac{3}{5} e^2}{4\pi\varepsilon_0 r}$$

For a nucleus having Z protons

$$w' = \frac{\frac{3}{5} Ze^2}{4\pi\varepsilon_0 r}$$

For a single-proton nucleus no work is done against Coulomb repulsion in assembling the nucleus. Thus, the true Coulomb energy term for a nucleus containing Z protons is $W - w$.

$$B_3 = -\left[\frac{\frac{3}{5} Z^2 e^2}{4\pi\varepsilon_0 r} - \frac{Z\frac{3}{5} e^2}{4\pi\varepsilon_0 r}\right]$$

i.e.

$$B_3 = -\frac{3}{5}\frac{Z(Z-1)e^2}{4\pi\varepsilon_0 r} \tag{2.7}$$

The negative sign indicates the repulsive term. As $r = r_0 A^{1/3}$, Eq. (2.7) can be written as

$$\boxed{B_3 = -a_c \frac{Z(Z-1)}{A^{1/3}}} \tag{2.8}$$

where a_c is constant.

Asymmetry Energy Term (B_4)

The asymmetry term reflects the stability of nuclei with the proton and neutron numbers being approximately equal. This is a term, which depends on the neutron excess ($N - Z$) in the nucleus and it decreases with the increasing nuclear binding energy. For very few nuclei of low Z, $N - Z = 0$ and are more stable compared to their neighbours, i.e. their binding energies are maximum. The reduction in binding energy for higher A nuclei is directly proportional to $(N - Z)^2$ or square of excess of neutrons and is inversely proportional to mass number. So, we can write

$$B_4 \propto \frac{(N-Z)^2}{A}$$

$$\boxed{B_4 = -a_a \frac{(A-2Z)^2}{A}} \tag{2.9}$$

As $A = N + Z$ and a_a is constant.

Pairing Energy Term (B_5)

So far we have all the terms in the binding energy have smooth variation with respect to N or Z or A. However, in the actual binding energy versus A curve, there are several discontinuities, particularly when N or Z becomes equal to 2, 4, 8, 20, 28, 50, 82 or 126. These values correspond to shell closure for N or Z. The nuclei having N or Z equal to one of these numbers have large binding energy. This fact did not appear in the liquid drop model, which does not consider intrinsic spin of the nucleons and the shell effects.

It is interesting to classify all the stable nuclei into four groups, first having even Z–even N, second even Z–odd N, third odd Z–even N and last having odd Z–odd N. This classification is shown in Table 2.1.

TABLE 2.1 Number of stable isotopes

Z	*N*	*Number of stable nuclei*
Even	Even	165
Even	Odd	55
Odd	Even	50
Odd	Odd	5

(the five stable odd *Z*–odd *N* nuclei are: $^{2}_{1}$H, $^{6}_{3}$Li, $^{10}_{5}$B, $^{14}_{7}$N, $^{180}_{73}$Ta)

From Table 2.1, it is clear that even *Z*–even *N* nuclei, being most stable, are most abundant. Accordingly, odd *Z*–odd *N* nuclei are least abundant and hence least stable. The remaining nuclei have intermediate stability. Therefore, the binding energy also depends upon whether the number of protons and neutrons are odd or even. This pairing effect was incorporated by putting

$$\boxed{B_5 = a_p A^{-3/4}} \tag{2.10}$$

where

a_p = 33.5 MeV for even–even nuclei

= 0 for odd–even (or odd *A*) nuclei

= –33.5 MeV for odd–odd nuclei

Substituting the values of B_1, B_2, B_3, B_4 and B_5 from Eqs. (2.5), (2.6), (2.8), (2.9) and (2.10) in Eq. (2.2), we get

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} + a_p A^{-3/4}$$

Substituting the value of *B* from the above equation in Eq. (2.1), we get the semiempirical mass formula as

$$\boxed{M = ZM_p + NM_n - a_v A + a_s A^{2/3} + a_c \frac{Z(Z-1)}{A^{1/3}} + a_a \frac{(A-2Z)^2}{A} - a_p A^{-3/4}} \tag{2.11}$$

The various constants found are

$$a_v = 15.5 \text{ MeV}$$
$$a_s = 16.8 \text{ MeV}$$
$$a_c = 0.7 \text{ MeV}$$
$$a_a = 23.0 \text{ MeV}$$
$$a_p \begin{cases} = 34 \text{ MeV} & \text{for even–even nuclei} \\ = 0 \text{ MeV} & \text{for odd } A \text{ nuclei} \\ = -34 \text{ MeV} & \text{for odd–odd nuclei} \end{cases}$$

The contribution of various terms for few cases are given (in MeV) in Table 2.2.

TABLE 2.2 Contribution of various terms of semiempirical mass formula for some isotopes

Term	$^{40}_{20}$Ca	$^{120}_{50}$Sn	$^{238}_{92}$U
Volume	+620	+1860	+3689
Surface	−196	−409	−645
Coulomb	−80	−358	−973
Asymmetry	0	−77	−282
Pairing	+2	+1	+0.6
Resultant BE	346	1017	1789.6
BE/n	8.65	8.48	7.52

The semiempirical mass formula reproduces masses of various nuclei quite accurately, but does not account for all the features of the nuclear binding energy.

2.2.2 Mass of Most Stable Isobar

Isobars are nuclides that have same mass number A. The semiempirical mass formula can predict the atomic number Z_0 of most stable isobar for given mass number A.

Neglecting 1 in comparison to Z in the Coulomb term and rewriting Eq. (2.11) as

$$M(Z, A) = aA + bZ + cZ^2 + \delta \tag{2.12}$$

where

$$a = M_n - \left[a_v - a_a - \frac{a_s}{A^{1/3}}\right]$$

$$b = -4a_a - (M_n - M_p)$$

$$c = \left(\frac{4a_a}{A} + \frac{a_c}{A^{1/3}}\right)$$

and

$$\delta = \mp a_p A^{-3/4}$$

Let us find the atomic number of most stable isotope for a given A. This can be calculated by taking the partial derivative of Eq. (2.12) with respect to Z keeping A as constant and equating the resultant equation to zero, i.e.

$$\left(\frac{\partial M}{\partial Z}\right)_A = b + 2cZ = 0$$

which gives

$$Z_0 = Z = -\frac{b}{2c}$$

where Z_0 is the atomic number of most stable isotope for given A. Substituting the values of b and c, we obtain

$$Z_0 = -\frac{-4a_a - (M_n - M_p)}{2\left[\frac{4a_a}{A} + \frac{a_c}{A^{1/3}}\right]}$$

Since all the quantities in this expression are known, atomic number for most stable isobar can be calculated.

2.2.3 Achievements of Liquid Drop Model

1. It predicts the atomic masses and binding energies of various nuclei accurately.
2. It predicts emission of a- and b-particles in radioactivity.
3. The theory of compound nucleus, which is based on this model, explains the basic features of the fission process.

2.2.4 Failures of Liquid Drop Model

1. It fails to explain the extra stability of certain nuclei, where the numbers of protons or neutrons in the nucleus are 2, 8, 20, 28, 50, 82 or 126 (these numbers are called magic numbers).
2. It fails to explain the measured magnetic moments of many nuclei.
3. It also fails to explain the spin of nuclei.
4. It is also not successful in explaining the excited states in most of the nuclei.
5. The agreement of semiempirical mass formula with experimentally observed masses and binding energies is poor for lighter nuclei compared to the heavy ones.

2.3 SHELL MODEL

Atomic theory based on the shell model has provided remarkable clarification of the complicated details of atomic structure. Nuclear physicist, therefore, attempted to use a similar theory to study nuclear structure. In the atomic shell model, we fill the shells with electrons in order of increasing energy consistent with the requirement of the Pauli principle. When we do so, one obtains an inert core of filled shells, containing 2, 10, 18, 36, 54 and 86 electrons (atomic numbers of inert gases) and some valence electrons; the atomic properties are determined primarily by the valence electrons. When we compare some measured properties of atomic system with the predictions of the model, one finds remarkable agreement. The same kind of effect has been observed in nuclei. Experimentally it was found that nuclei that have 2, 8, 20, 28, 50, 82 and 126 nucleons (protons or neutrons), called magic numbers, are more abundant than other nuclei.

However, there exist several significant differences between atomic and nuclear cases. In the atomic case, the potential is provided by the Coulomb field of the nucleus; the orbits are generated by the external agent i.e. interaction between electrons and nucleus. We can solve the Schrödinger equation for this potential and calculate the energies of the sub-shells into which electrons can then be placed. In case of nucleus, there is no such external agent, the nucleons move in a potential which is not well defined that they themselves create.

Another appealing aspect of atomic shell theory is the existence of spatial orbits. It is often very useful to describe atomic properties in terms of spatial orbits of the electrons. The electrons can move in those orbits relatively free of collisions with other electrons. Nucleons which have a mass about 2000 times larger than that of electrons have a diameter comparable to the size of the nucleus, which is about 10^5 times smaller than that of an atom. How can we regard the nucleons as moving in well-defined orbits when a single nucleon can make many collisions during each orbit?

All these observations tempted nuclear physicists (Barlet, Guggenheimer et al.) to devise an independent particle model formally called the shell model. A shell structure means that nucleons move freely inside the nucleus similar to the electron motion in atom. This approach could explain the existence of first few magic numbers. However, physicist lost interest in this model till 1948 due to its failure to explain higher magic numbers.

In 1948, M.G. Mayer in USA brought together a considerable amount of convincing information showing the evidence for the closed shells, which led to the development of nuclear shell model which could explain all the magic numbers, namely 2, 8, 20, 28, 50, 82 and 126, which apparently represent closed shells in the nucleus. Some of the main aspects of this evidence based on the study of stable nuclei are as under:

1. Binding energy per nucleon vs. A curve. If we plot binding energy per nucleon versus A curve, it shows that binding energy suddenly increases when the number of nucleons is either 2, 8, 20, 28, 50, 82 or 126 indicating that these nuclei are exceptionally stable.
2. Number of stable isotopes. Relative stabilities of different elements are also indicated by the number of stable isotopes per element.

1.

$_{19}K = 3$	$_{20}Ca = 6$	$_{21}Sc = 1$
$_{49}In = 2$	$_{50}Sn = 10$	$_{51}Sb = 2$
$_{81}Tl = 2$	$_{82}Pb = 4$	$_{83}Bi = 1$

It is clear that number of stable isotopes for z = 20, 50 and 82 are much larger compared to neighbouring isotopes.

3. Number of stable isotones. The numbers of stable isotones for N = 19, 20, 21; 49, 50, 51 and 81, 82, 83 are shown in Table 2.3.

TABLE 2.3 Number of stable isotones around different magic numbers

N	*Stable isotones*	*N*	*Stable isotones*	*N*	*Stable isotones*
19	0	49	1	81	1
20	5	50	6	82	7
21	1	51	1	83	1

It is clear from the above table that the numbers of stable isotones for N = 20, 50 and 82 are much larger as compared to neighbouring stable isotones.

4. A table of relative abundances of nuclei compiled from data on the composition of earth, sun, stars and meteorites shows pronounced peaks at

^{16}O	($N = Z = 8$)
^{40}Ca	($N = Z = 20$)
^{118}Sn	($Z = 50$)
^{88}Sr, ^{89}Y, ^{90}Zr	($N = 50$)
^{138}Ba, ^{139}La, ^{140}Ce	($N = 82$)
^{208}Pb	($Z = 82$, $N = 126$)

5. Binding energy of next neutron after a magic number is small. The separation energy of the last neutron for N = 7, 8, 9; 19, 20, 21 and 27, 28 29 is shown in Table 2.4.

TABLE 2.4 Binding energy of the last neutron around magic numbers

Nucleus	*N*	S_n (MeV)	*Nucleus*	*N*	S_n (MeV)	*Nucleus*	*N*	S_n (MeV)
^{15}O	7	13.2	^{39}Ca	19	13.3	^{47}Ca	27	7.3
^{16}O	8	15.7	^{40}Ca	20	15.7	^{48}Ca	28	9.9
^{17}O	9	4.14	^{41}Ca	21	8.4	^{49}Ca	29	5.1

From the above table it is clear that for neutron numbers = 9, 21 and 29, the separation energy of the last neutron suddenly decreases as compared to the case, when neutron numbers are 8, 20 and 28.

6. It is found that some isotopes are spontaneous neutron emitters. They are:

$^{17}_{8}O_{9}$, $^{81}_{36}Kr_{51}$, $^{137}_{54}Xe_{83}$, $^{89}_{36}Kr_{53}$

7. Decay product of radioactive series:

Thorium $^{232}_{90}Th_{142}$	1.4×10^{10} y($4n$ series)	$^{208}_{82}Pb_{126}$
Uranium $^{238}_{92}U_{148}$	4.51×10^{9} y($4n + 2$ series)	$^{206}_{82}Pb_{124}$
Actinium $^{235}_{92}U_{143}$	7.07×10^{8} y($4n + 3$ series)	$^{207}_{82}Pb_{125}$
Neptunium $^{237}_{93}Np_{144}$	2.25×10^{6} y($4n + 1$ series)	$^{209}_{83}Bi_{126}$

The end product of each series ends in N or Z equal to either 82 or 126.

8. Neutron absorption cross-section s, the probability of absorption of neutron by the nucleus is small for nuclides containing magic number of neutrons

N	19	20	21	49	50	51
s	2.0	0.41	12.0	19.0	0.65	6.4

9. a-decay energies are rather smooth functions of A for a given Z, but it shows striking discontinuities at $N = 126$ or $Z = 82$, the energy of a-particles increases. For example,

$$^{212}_{84}Po_{128} \rightarrow {}^{208}_{82}Pb_{126} + a(9\text{ MeV})$$ Half-life $t_{1/2} = 3 \times 10^{-7}$s

where $N = 126$, $Z = 82$

$$^{210}_{84}Po_{126} \rightarrow {}^{206}_{82}Pb_{124} + a(6\text{ MeV})$$ $t_{1/2} = 3 \times 10^{-7}$s

where $N = 126$, $Z = 82$

10. Similar behaviour is exhibited by b-emitters.

11. The electric quadrupole moment measures the departure of nuclear charge distribution from sphericity. This departure is a measure of nuclear deformation. A spherical nucleus has no or nearly zero quadrupole moments. It has been found that nuclei with proton or neutron number equal to one of the magic numbers are spherical in nature, i.e. all the three axes x, y and z are equal like that of a tennis

ball. For such nuclei the quadrupole moment is either zero or nearly zero which is also observed experimentally.

However, in some nuclei, out of three, two axes are equal. In the case, where the unequal axis is shorter than the others, the nucleus has somewhat of a pumpkin shape, it is called *oblate*. Extreme case is that of a Hamburger as shown in Figure 1.7. In the other case where the third unequal axis is longer than the other two, the nucleus has somewhat of a football shape, it is called *prolate*. Extreme case is that of a cigar or Hot Dog as shown in Figure 1.6. In general, unequal axis differs in length by about 20%. However, in lighter nuclei, deformations are more. For example, in ^{24}Mg, all the three axes are unequal.

Thus all the facts given above show that magic numbers 2, 8, 20, 28, 50, 82 and 126 correspond to closed shells. The nuclei having any one of these magic number of protons or neutrons or both show more stability than the other nuclei.

Basic Assumptions of the Shell Model

Calculations similar to atoms were also performed for the nucleus also. Following assumptions were made for these calculations:

- Nucleons in a nucleus move independently in a common (mean) potential determined by the average motion of all the other nucleons.
- Protons and neutrons separately fill levels in the nucleus.
- Most of the nucleons are paired and a pair of nucleons contributes zero spin and zero magnetic moment. The paired nucleons thus form an inert core.
- The properties of odd A nuclei are characterized by the unpaired nucleon and odd–odd nuclei by the unpaired proton and neutron.

These assumptions indicate that the nucleus might have a shell structure. It means that nucleons moving in different shells inside the nucleus do not suffer any collisions similar to electrons in different shells in the atom. This assumption is apparently not acceptable as the nucleons have almost the same size as that of the nucleus. So obviously a question arises, why do not so many nucleons moving inside the nucleus suffer any collisions? How can

we regard the nucleons as moving in well-defined orbits when a single nucleon can make many collisions during each orbit?

The answer to this question comes from Pauli's exclusion principle. Consider in a heavy nucleus, a collision between two nucleons in a state near the very bottom of the potential well. When the nucleons collide, they transfer energy to one another, but if all of the energy levels are filled up to the level of valence nucleon, there is no way for one of the nucleons to gain energy except to move up to the valence level. The other levels near the original level are filled and cannot accept an additional nucleon. Such a transfer from a low-lying level to the higher-lying level requires more energy than the nucleons are likely to transfer in a collision. Thus, the collisions cannot occur and the nucleons can indeed orbit as if they were transparent to one another!

The first step in developing the shell model is the choice of the potential. Different forms of potential $V(r)$ have been employed in order to obtain the required magic numbers. In the following, we consider two potentials to solve the Schrödinger equation.

2.3.1 The Square Well Potential

The problem can be mathematically simplified, if we assume a potential well with infinite walls as

$V(r) = -V_0 \; r < r_0$

$= 0 \; r > r_0$

The shape of the finite square well potential is shown in Figure 2.1.

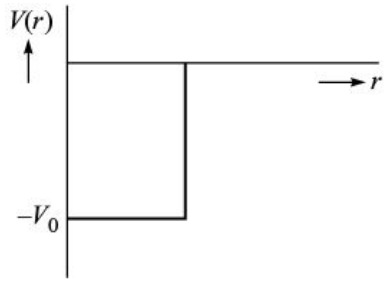


Figure 2.1 Square well potential. –V0 is the depth of the well.

If we solve Schrödinger equation for square well potential, we get the following sequence of levels:

- 1*s*
- 1*p*
- 1*d*
- 2*s*
- 1*f*
- 2*p*
- 1*g*
- 2*d*
- 1*h*
- 3*s*
- 2*f*
- 1*i*
- 3*p*
- 2*g*

and so on, where *s*, *p*, *d*, *f*, *g*, *h*, *i*, …, etc. stand for usual spectroscopic notation □ = 0, 1, 2, 3, 4, 5, 6, …, respectively.

Because of the two different spin orientation of the nucleon, a level can contain (2□ + 1) protons or neutrons. For example, number of nucleons in 1*s* (□ = 0) shell will be 2(2 □ 0 + 1) = 2 and number of nucleons in 1*f* (□ = 3) shell will be 2(2 □ 3 + 1) = 14. This model predicts the shell closures at nucleon number 2, 8, 18, 20, 34, 40, 58, etc. as shown in Table 2.5.

TABLE 2.5 Nuclear levels and magic numbers predicted by square well potential

Level	*Number of nucleons*	*Magic numbers*
1*s*	2	2
1*p*	6	8
1*d*	10	18
2*s*	2	20
1*f*	14	34
2*p*	6	40
1*g*	18	58
2*d*	10	68
1*h*	22	90
3*s*	2	92
2*f*	14	106
1*i*	26	132
3*p*	6	138

The numbers shown in column 3 of the above table are not the observed magic numbers. The sequence of levels for the square well potential is shown in Figure 2.2.

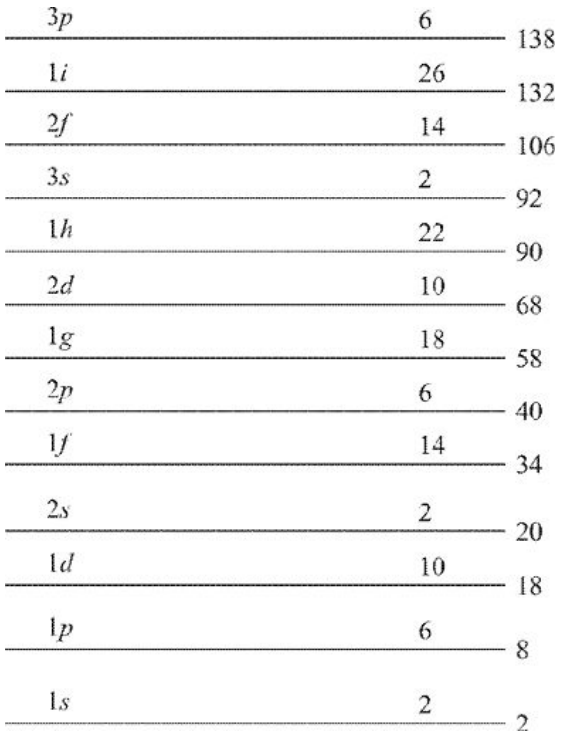


Figure 2.2 Sequence of levels of the square well potential.

The level sequence for square well potential can be remembered in the following way. First in a vertical column write the level sequence 1*s*, 1*p*, 1*d*,

1*f*, 1*g*, 1*h*, 1*i*, etc. as shown in Figure 2.3. Then leave two vertical spaces as blank and again write the level sequence 2*s*, 2*p*, 2*d*, 2*f*, 2*g*, etc. Again leave two vertical blank spaces and write the level sequence 3*s*, 3*p*, 3*d*, 3*f*, etc. as shown in Figure 2.3. In this sequence, 3*s* level will shift between 1*h* and 2*f*, similarly, 1*i* level will shift between 2*f* and 3*p* as indicated by arrow in the Figure 2.3. The resulting sequence will look as shown in Figure 2.4.

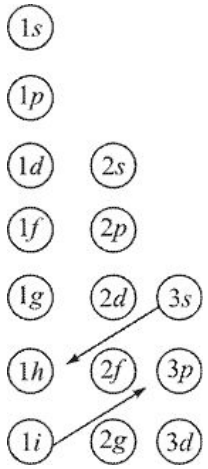


Figure 2.3 Way to remember the square well potential levels.

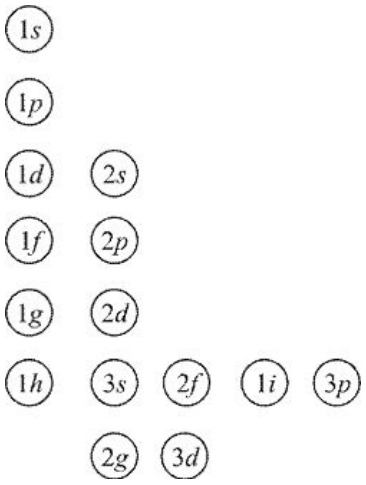


Figure 2.4 The resulting sequence is the sequence of square well potential levels.

Now, starting from the top and move horizontally from left to right, the first level is 1*s*, second 1*p*, third 1*d*, fourth 2*s*, fifth 1*f*, and so on.

2.3.2 The Harmonic Oscillator Potential

The shape of this potential is shown in Figure 2.5.

$$V(r) = -V_0 + \frac{1}{2}Kr^2 \qquad (2.13)$$

This potential and the square well potential provide two contrasting view points. The square well has infinite sharp edges. The harmonic oscillator potential diminishes steadily at the edges.

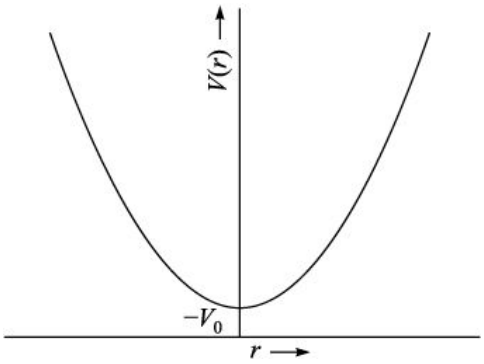


Figure 2.5 Harmonic oscillator potential.

Again solving Schrödinger equation for harmonic oscillator potential, we get the following sequence of levels as shown in first column of Table 2.6. The first level is 1*s*, second is 1*p*, third level contains two sub-shells 2*s*, 1*d*, having same energy, fourth level again contains two sub-shells, 2*p* and 1*f*, and so on.

TABLE 2.6 Nuclear levels and magic numbers predicted by harmonic oscillator potential

Level	*No. of nucleons in various levels*	*Magic number*
1*s*	2	2
1*p*	6	8
2*s*, 1*d*	2 + 10 = 12	20
2*p*, 1*f*	6 + 14 = 20	40
3*s*, 2*d*, 1*g*	2 + 10 + 18 = 30	70
3*p*, 2*f*, 1*h*	6 + 14 + 22 = 42	112
4*s*, 3*d*, 2*g*, 1*i*	2 + 10 + 18 + 26 = 56	168

In this case also each sub-shell contains 2(2□ + 1) protons or neutrons. For example, in fourth shell, we have two sub-shells, 2*p* and 1*f*. For 2*p* (□ = 1), it contains 2(2 □ 1 + 1) = 6 nucleons and 2*f* (□ = 3), it contains 2(2 □ 3 + 1) = 14 nucleons. Total number of nucleons in fourth shell = 6 + 14 = 20. These numbers are shown in the third column of the Table 2.6.

This level sequence again does not reproduce experimentally observed magic numbers. The sequence of harmonic oscillator levels is shown in

Figure 2.6. The levels are equally spaced. Harmonic oscillator level sequence can also be remembered in almost similar way as that of square well potential as shown below.

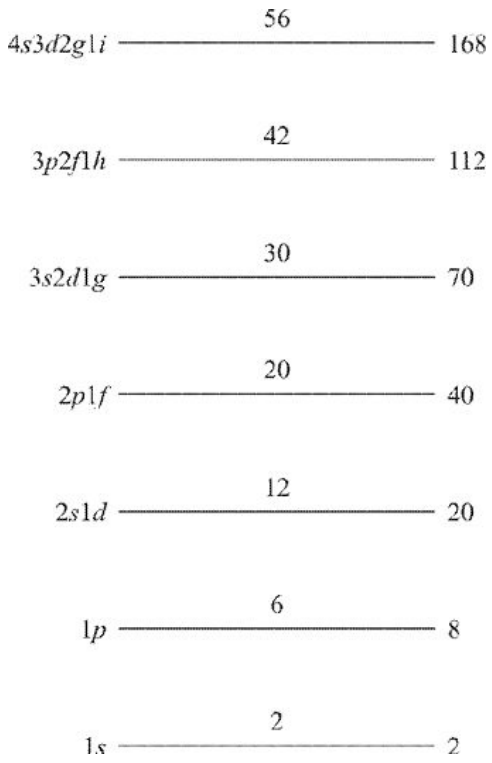


Figure 2.6 Level sequence as obtained for the harmonic oscillator levels.

As shown in Figure 2.7, write in the first vertical column the level sequence 1*s*, 1*p*, 1*d*, 1*f*, 1*g*, 1*h*, 1*i*, etc. Then leave two vertical spaces as blank and again write the level sequence 2*s*, 2*p*, 2*d*, 2*f*, 2*g*, etc. Again leave two vertical blank spaces and write the level sequence 3*s*, 3*p*, 3*d*, 3*f*, etc. Now, these sequences read horizontally are the harmonic oscillator levels. For example, first level is 1*s*, second is 1*p*, third is 1*d*, 2*s* and so on.

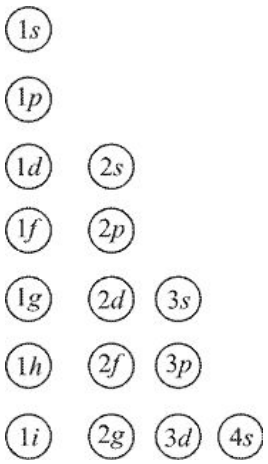


Figure 2.7 Way to remember the harmonic oscillator levels.

The other potential, which is a compromise between square well and harmonic oscillator potential, is

$$V(r) = \frac{-V_0}{1 + e^{\frac{R-r}{d}}} \tag{2.14}$$

This potential is known as Woods–Saxon potential. In this equation $d = 0.524$ fm, R is the mean nuclear radius and $r = r_0 A^{1/3}$. Unlike square well potential, the Woods–Saxon potential does not have any sharp edges at all. The harmonic oscillator potential also does not have any edges. The shape of this potential is shown in Figure 2.8. This potential closely approximates the nuclear charge and matter distribution, falling smoothly to zero beyond the mean radius R. When the Schrödinger equation was solved for this potential, it predicted 2, 8, 20, 40, 58, 92, 112 as magic numbers. We again get the magic numbers 2, 8, and 20, but the higher magic numbers do not emerge from the calculations.

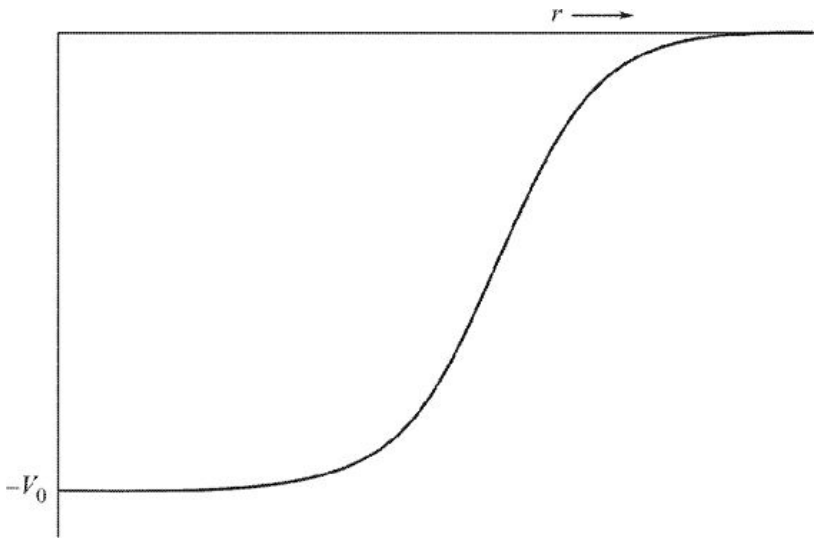


Figure 2.8 Wood–Saxon potential.

2.3.3 Spin-Orbit Coupling

A way out of this difficulty, which proved to be remarkably successful was proposed independently in 1949 by M.G. Mayer in USA and O. Haxel, J.H.D. Jensen and H.E. Suess
in Germany. We have seen in Chapter 1 that each nucleon has a spin angular momentum
$|\vec{s}| = \sqrt{s(s+1)\hbar}$ and orbital angular momentum $|\vec{\ell}| = \sqrt{\ell(\ell+1)\hbar}$. It was proposed that there is a strong coupling between the orbital and spin angular momentum of each individual nucleon; referred as spin-orbit coupling. As a result of the spin-orbit coupling, the nucleon energy level for a given value □ of the orbital quantum number (except □ = 0) splits into two sub-levels, characterized by total angular momentum quantum number j = □ + 1/2 and j = □ – 1/2 corresponding to spin components of +1/2 and –1/2 respectively. The sign of this term is chosen in such a way that □ + 1/2 level goes down in energy whereas □ – 1/2 goes up. Further, the total splitting is proportional to □ and becomes so large that for a given n, the level with largest □ value slides down to energy as low as those of the multiplet with quantum number $n - 1$.

$1s_{1/2}$	2
$1p_{3/2}$, $1p_{1/2}$	8
$1d_{5/2}$, $2s_{1/2}$, $1d_{3/2}$	20

$1f_{7/2}$	28
$2p_{3/2}$, $1f_{5/2}$, $2p_{1/2}$, $1g_{9/2}$	50
$1g_{7/2}$, $2d_{5/2}$, $2d_{3/2}$, $3s_{1/2}$, $1h_{11/2}$	82
$1h_{9/2}$, $2f_{7/2}$, $2f_{5/2}$, $3p_{3/2}$, $3p_{1/2}$, $1i_{13/2}$	126
$2g_{9/2}$, $3d_{5/2}$, $1i_{11/2}$, $2g_{7/2}$, $4s_{1/2}$, $3d_{3/2}$, $1j_{15/2}$	184

The sequence of these levels is shown in Figure 2.9.

2.3.4 Predictions of the Shell Model

1. Magic numbers.
2. Even–even nuclei have ground state angular momentum or spin 0. There is no known exception to this rule.
3. In odd A nuclei the spin will be determined by the last unpaired particle. For example, in $^{13}_{6}C_{7}$ and $^{13}_{7}N_{6}$ the levels fill as under

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^1$$

Thus, in $^{13}_{6}C_{7}$ unpaired neutron is in $1p_{1/2}$ shell and, therefore, nucleus has a spin 1/2, whereas in $^{13}_{7}N_{6}$ last unpaired proton is also in $1p_{1/2}$ shell so its spin should also be 1/2. This is indeed observed experimentally.

Similarly, in $^{17}_{8}O_{9}$ and $^{17}_{9}F_{8}$ the filling of levels will be

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^2 \mid (1d_{5/2})^1$$

the predicted spin is 5/2, which is also experimentally observed. Similarly, for $^{33}_{16}S_{17}$

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^2 \mid (1d_{5/2})^6 (2s_{1/2})^2 (1d_{3/2})^1$$

predicted and observed spin is 3/2. However, in $^{75}_{33}As_{42}$ and $^{61}_{28}Ni_{33}$

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^2 \mid (1d_{5/2})^6 (2s_{1/2})^2 (1d_{3/2})^4 \mid (1f_{7/2})^8 \mid (2p_{3/2})^4 (1f_{5/2})^1$$

predicted spin is 5/2, whereas observed is 3/2, for both these nuclei.
Similar kind of exceptions has been observed for neutron numbers 57, 59 and 61 also.

One would also expect that for a high atomic mass number A, there will be many stable nuclei with spin 11/2 corresponding to an odd nucleon in the $1h_{11/2}$ state and similarly there should be many stable nuclei with spin 13/2 corresponding to an odd nucleon in the $1i_{13/2}$ state, but not even a single nucleus has ever been observed with ground state spin of 11/2 or 13/2. Many such exceptions have been eliminated by modifying the rules and stating that if the high spin shell (say $1f_{5/2}$) comes after low spin shell ($2p_{3/2}$), the high spin shell fills faster, pairing its particles before the low spin shell can be filled completely. According to this rule, we may write for ^{75}As and ^{61}Ni.

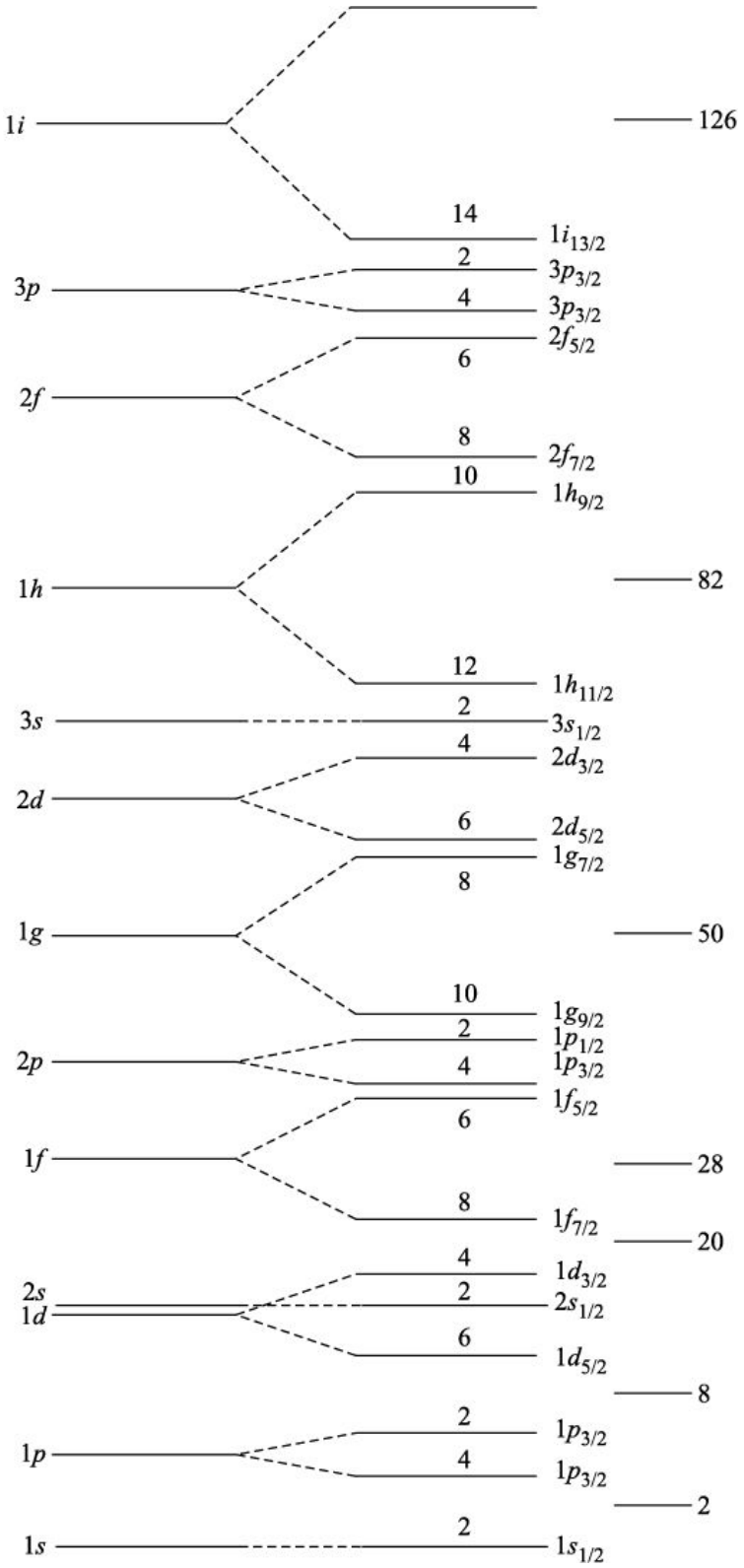


Figure 2.9 Level scheme due to spin-orbit coupling.

$$(1s_{1/2})^2 \mid (1p_{3/2})^4\,(1p_{1/2})^2 \mid (1d_{5/2})^6\,(2s_{1/2})^2\,(1d_{3/2})^4 \mid (1f_{7/2})^8 \mid (2p_{3/2})^3\,(1f_{5/2})^2$$

giving spin as 3/2 for both these nuclei.

We can say that there is a strong tendency for particles to form pairs in higher □ states even at some expense of energy. This can be put into the model in the form of pairing potential, which gives paired nucleons a lower energy than unpaired ones, and which increases with increasing □.

The higher angular momentum states are usually formed in pairs. Thus, a level $1h_{11/2}$ may be filled in pairs while the odd nucleon goes to $3s_{1/2}$ or $2d_{3/2}$ shell. For example, the measured spin of $^{137}_{56}\mathrm{Ba}_{81}$ is 3/2, while the one predicted by the shell model is 11/2.

$(1s_{1/2})^2 \mid (1p_{3/2})^4 \ (1p_{1/2})^2 \mid (1d_{5/2})^6 \ (2s_{1/2})^2 \ (1d_{3/2})^4 \mid (1f_{7/2})^8 \mid$
$(2p_{3/2})^4 \, (1f_{5/2})^6 \, (2p_{1/2})^2 \, (1g_{9/2})^{10}$
$\mid (1g_{7/2})^8 \, (2d_{5/2})^6 \, (2d_{3/2})^4 \, (3s_{1/2})^2 \, (1h_{11/2})^{11} \mid$

predicts spin as 11/2.

Due to pairing in higher l states, the alternative arrangement of nucleons is as under

$(1s_{1/2})^2 \mid (1p_{3/2})^4 \ (1p_{1/2})^2 \mid (1d_{5/2})^6 \ (2s_{1/2})^2 \ (1d_{3/2})^4 \mid (1f_{7/2})^8 \mid$
$(2p_{3/2})^4 \, (1f_{5/2})^6 \, (2p_{1/2})^2 \, (1g_{9/2})^{10} \mid$
$(1g_{7/2})^8 \, (2d_{5/2})^6 \, (2d_{3/2})^3 \, (3s_{1/2})^2 \, (1h_{11/2})^{12} \mid$

This predicts spin as 3/2. A discrepancy occurs at $^{19}_{9}\mathrm{F}_{10}$, which according to the shell model should have spin of 5/2.

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 \, (1p_{1/2})^2 \mid (1d_{5/2})^1 \, (2s_{1/2})^0 \, (1d_{3/2})^0$$

Experimentally spin is 1/2. The odd proton goes to $2s_{1/2}$ state instead of $1d_{5/2}$ state. This discrepancy may be explained as the result of coupling between the nucleons outside the closed shell, i.e. the two neutrons each having spin 5/2 and a proton also having spin 5/2. Another discrepancy is for $^{23}\mathrm{Na}$, which has a spin of 3/2 while the one predicted by the shell model is 5/2.

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 \, (1p_{1/2})^2 \mid (1d_{5/2})^3 \, (2s_{1/2})^0 \, (1d_{3/2})^0$$

Coupling between 3 protons in $1d_{5/2}$ shell gives spin of 3/2.

Finally, the parity of the system is given by $(-1)^{\square}$, where $\square$ is the orbital quantum number of the last odd nucleon. For a nucleon in a state *s*, *d*, *g*, …

corresponding to □ = 0, 2, 4, … the parity is even (+), while for the states *p*, *f*, *h*, … corresponding to □ = 1, 3, 5, … the parity is odd (–).

2.3.5 Achievements of the Shell Model

1. It explains the ground state spin and parities of all even–even nuclei without any exception.
2. It explains the ground state spin and parities of most of odd *A* (even–odd or odd–even) nuclei.
3. It also explains the spin and parities of odd–odd nuclei.
4. It explains the extra stability of magic nuclei.
5. It also explains the qualitative features of magnetic dipole and electric quadrupole moments of different nuclei.
6. It is also able to explain many other properties, like nuclear isomerism of different nuclei.

2.3.6 Failures of Shell Model

1. Shell model fails to explain spin values for certain nuclei.
2. Shell model is unable to explain the energy of first excited states in even–even nuclei.
3. It is unable to explain magnetic moments of some nuclei.
4. This model is also unable to explain quadrupole moments of many nuclei.
5. Shell model is also unable to explain the ground states of odd *A* nuclei in the mass region 150 □ *A* □ 190 and *A* □ 220.

2.4 FERMI GAS MODEL

The semiempirical binding energy formula is based on treating the nucleus like a liquid drop. Such an analogy is an oversimplification and nucleus has many properties that can be explained more simply in terms of independent particle behaviour rather than in terms of the strong interaction picture implied by the liquid drop model. The most primitive independent particle model is obtained if the nucleus is treated as a degenerate Fermi gas of nucleons. The nucleons are assumed to move freely except for effects of the

exclusion principle, throughout a sphere of radius $R = R_0 A^{1/3}$, $R_0 = 1.2$ fm. The situation is represented in Figure 2.10 by two wells, one for neutrons and other for protons. Free neutrons and free protons, far away from the walls, have the same energy, and the zero level for the two wells is the same. The two potential wells though have slightly different shapes, mainly because of the Coulomb part, the well for protons is less deep because of the Coulomb potential by an amount E_C and externally the $\frac{1}{r}$ dependence of the Coulomb potential extends the range. Protons trying to enter the nucleus from the outside are repelled by the nuclear charge; they must either tunnel through the barrier explained in Chapter 3 or have enough energy to pass over the barrier.

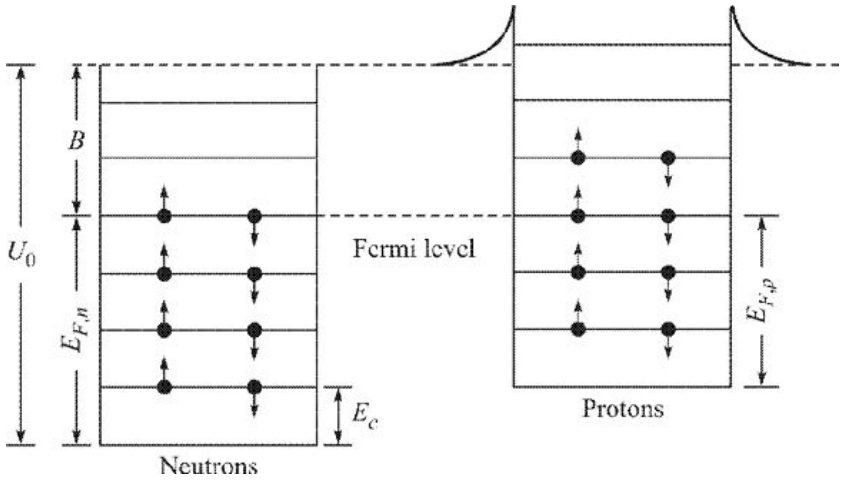


Figure 2.10 Protons–neutrons levels in Fermi model of the nucleus. The two potential wells have slightly different shapes because of the Coulomb repulsion of protons.

The wells contain a finite number of levels; each level can be occupied by two nucleons, one with spin up and other with spin down. It is assumed that the nuclear temperature is so low that the nucleons occupy the lowest states available to them. The term degenerate Fermi gas describes such a situation. The nucleons populate all states up to a maximum kinetic energy equal to the Fermi energy E_F.

From the density of the states obtained from a free Fermi gas confined to stay within a volume V, the nuclear volume, the number of particle states are given by

$$n = \frac{2V}{(2\pi\hbar)^3} \int_0^{P_F} d^3P \tag{2.15}$$

or

$$n = \frac{VP_F^3}{3\pi^2\hbar^3}$$

and the Fermi momentum P_F is given by

$$P_F = \hbar\left(3\pi^2 \frac{n}{V}\right)^{1/3} = h\left(\frac{3}{8\pi}\frac{n}{V}\right)^{1/3} \tag{2.16}$$

Let us calculate the depth of the potential well for Z protons and N neutrons. We can write

$$P_{F,n} = \frac{\hbar}{r_0}\left(\frac{9\pi N}{4A}\right)^{1/3}$$

$$P_{F,p} = \frac{\hbar}{r_0}\left(\frac{9\pi Z}{4A}\right)^{1/3}$$

for the neutron and proton Fermi momentum respectively. We have used the relations

$V = 4/3\,p\,r_0^3 A.$ A simple estimate of the Fermi momentum can be reached by considering

$$N = Z = \frac{A}{2},$$

resulting in

$$P_{F,n} \approx P_{F,p} \approx \frac{\hbar}{r_0}\left(\frac{9\pi}{8}\right)^{1/3}$$

Using the value $\hbar c$ = 197 MeV fm, this becomes

$$P_{F,n} \approx P_{F,p} \approx \frac{297}{r_0}\ \text{MeV/c} \tag{2.17}$$

The corresponding Fermi kinetic energy with r_0 = 1.2 fm becomes

$$\in_F = \frac{P_{F,n}^2}{2m} \approx \frac{P_{F,p}^2}{2m} \approx 33\ \text{MeV} \tag{2.18}$$

This energy corresponds to the kinetic energy of the highest occupied orbit (smallest binding energy). Given average binding energy □8 MeV, we can

make a good estimate of the nuclear well depth □33 + 8 = 41 MeV, which agrees well with the experimental value.

The Fermi gas model gave some information about nuclear excited states even at low excitation energies observed in radioactive decay.

2.5 COLLECTIVE MODEL

The successes of the these models presented here led to a serious dilemma, whereas the liquid drop model can account for the behaviour of nucleus as a whole, shell model indicates the individual and nearly independent behaviour of nucleons. Combining the two aspects together, a new model called Collective Model was developed. In addition, this new model gave good results for magnetic moments, excited states, etc. of some nuclei. Giving more details of this model is beyond the scope of this book.

NUMERICAL PROBLEMS

Section 2.2

Solved Problems

1. Use the semiempirical mass formula to calculate the binding energy of $^{40}_{20}$Ca. What is the percentage discrepancy between this value and the actual value?

Solution: Binding energy = $[M(Z, N) - Zm_p - Nm_n]$ □ 931.49 MeV

For $^{40}_{20}$Ca, actual binding energy

$= [M(20, 20) - 20m_p - 20m_n]$ □ 931.49 MeV

$= (39.962591 - 20$ □ $1.007825 - 20$ □ $1.008665)$ □ 931.49 MeV

$= -342.05$ MeV

Binding energy as per semiempirical mass formula

$$B = a_v - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} \pm a_p A^{-3/4}$$

Here, $A = 40$, $Z = 20$, $N = 20$, $a_v = 15.5$ MeV, $a_s = 16.8$ MeV, $a_c = 0.7$ MeV,

$a_a = 23.0$ MeV and $a_p = 34.0$ MeV

Substituting various values, we get

$$B = 40 \square 15.5 - 16.8 \square 40^{2/3} - 0.7 \square \frac{20 \times 19}{40^{1/3}} - 23.0 \frac{40 - 2 \times 20}{40} - 30 \square 40^{-3/4}$$

$$= 620 - 196.4939 - 77.77887 - 2.1376$$

$$= 343.59 \text{ MeV}$$

Therefore, percentage discrepancy between the actual and semiempirical mass formula value

$$= \frac{343.59 - 342.05}{342.05}$$

$$\approx 0.5\%$$

2. Coulomb energy is given as

$$E_c = \frac{3}{5} \frac{Z(Z-1)e^2}{4\pi\varepsilon_0 R}$$

For mirror nuclei such as $^{15}_{7}\text{N}$ and $^{15}_{8}\text{O}$, the difference in masses $\square m$ is due to difference in Coulomb energy and difference between proton and neutron mass. Show whether this value of $\square m$ agrees with the actual value.

Solution: Given:

$$E_c = \frac{3}{5}\frac{Z(Z-1)\,e^2}{4\pi\varepsilon_0 R}$$

or

$$E_c = a_c \frac{Z(Z-1)}{A^{1/3}},$$

where $$a_c = \frac{3e^2}{5\times 4\pi\varepsilon_0 r_0^{1/3}}$$

$$= 0.7 \text{ MeV}$$

For $^{15}_{7}$N, Coulomb energy contribution is

$$E_c = \frac{0.7\times 7\times 6}{15^{1/3}}$$

$$= 11.92 \text{ MeV}$$

For $^{15}_{8}$O, Coulomb energy contribution is

$$E_c = \frac{0.7\times 8\times 7}{15^{1/3}}$$

$$= 15.895 \text{ MeV}$$

Difference in both the Coulomb energy contribution = 15.895 – 11.92 = 3.975 MeV

Difference in the masses of $^{15}_{8}$O and $^{15}_{7}$N is simply the difference in the masses of hydrogen and neutron. Therefore,

$$m_O - m_N = (m_H - m_n) \times 931.49 \text{ MeV}$$

$$= (1.007276 - 1.008665) \times 931.49 \text{ MeV}$$

$$= -1.2938 \text{ MeV}$$

Therefore, according to the problem, difference in Coulomb energy and difference between proton and neutron mass is given by

$$= -1.2938 + 3.975 = 2.681 \text{ MeV}$$

Actual difference in the masses of $^{15}_{8}$O and $^{15}_{7}$N

$$= 15.003070 - 15.000108$$

$$= 0.002962 \text{ amu}$$

$$= 0.002962 \times 931.49$$

$$= 2.759 \text{ MeV}$$

Therefore, for mirror nuclei, the difference in masses Δm is due to difference in Coulomb energy and difference between proton and neutron

mass.

3. Find the energy needed to remove a neutron from ^{81}Kr, ^{82}Kr and ^{83}Kr.

Solution: Separation energy of neutron for ^{81}Kr = [*BE* of ^{81}Kr – *BE* of ^{80}Kr]

Binding energy = $[Zm_p + Nm_n - M(Z, N)]$ □ 931.49 MeV

For ^{81}Kr, binding energy = (36 □ 1.007825 + 45 □ 1.008665 – 80.91661) □ 931.49 MeV

= 703.28 MeV

For ^{80}Kr, binding energy = (36 □ 1.007825 + 44 □ 1.008665 – 80.91661) □ 931.49 MeV

= 695.50 MeV

Therefore, separation energy of neutron for ^{81}Kr = 703.27 – 695.49 = 7.78 MeV

Separation energy of neutron for ^{82}Kr = [*BE* of ^{82}Kr – *BE* of ^{81}Kr]

For ^{81}Kr, binding energy = (36 □ 1.007825 + 46 □ 1.008665 – 81.913482) □ 931.49 MeV

= 714.28 MeV

Therefore, separation energy of neutron for ^{82}Kr = 714.28 – 703.28 = 11.00 MeV

Separation energy of neutron for ^{83}Kr = [*BE* of ^{83}Kr – *BE* of ^{82}Kr]

For ^{83}Kr, binding energy = (36 □ 1.007825 + 47 □ 1.008665 – 82.914134) □ 931.49 MeV

= 721.74 MeV

Therefore, separation energy of neutron for ^{83}Kr = 721.74 – 714.28 = 7.46 MeV

4. Which isobars of $A = 75$ does the liquid drop model suggest to be the most stable nucleus?

Solution: According to liquid model, binding energy is given by

$$B = a_v - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} \pm a_p A^{-3/4}$$

Here, a_v = 15.5 MeV, a_S = 16.8 MeV, a_c = 0.7 MeV, a_a = 23.0 MeV and a_p = 34.0 MeV

In order to find most stable isotope for A = 75, we take the partial derivative of B with respect to Z and equate that to zero.

$$\frac{\partial B}{\partial Z} = -a_c \frac{2Z-1}{A^{1/3}} + 4a_a \frac{A-2Z}{A} = 0$$

Substituting various values

$$= -\frac{0.7}{75^{1/3}}(2Z-1) + \frac{4\times 23}{75}(75-2Z) = 0$$

which gives

$$Z = 32.89 \approx 33$$

Therefore, according to liquid drop model, most stable isotope for A = 75 is with Z = 33.

5. Using the liquid drop model, find the most stable isobars for A = 27, A = 118 and A = 238.

Solution: Binding energy is given by

$$B = a_v A - a_s A^{2/3} - a_c \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(A-2Z)^2}{A} \pm a_p A^{-3/4}$$

For most stable isobar, $\frac{\partial B}{\partial Z} = 0$

or $$\frac{\partial B}{\partial Z} = \frac{a_c}{A^{1/3}}(2Z-1) + \frac{4a_a}{A}(A-2Z) = 0$$

which gives

$$Z = \frac{4a_a + \frac{a_c}{A^{1/3}}}{\frac{2a_c}{A^{1/3}} + \frac{8a_a}{A}}$$

Substituting a_c = 0.7 MeV and a_a = 23.0 MeV and A = 27, we get

$$Z = \frac{4 \times 23 + \dfrac{0.7}{27^{1/3}}}{\dfrac{2 \times 0.7}{27^{1/3}} + \dfrac{8 \times 23}{27}} \approx 12.6$$

or $Z = 13$

Similarly, for $A = 118$, we get $Z = 49.8$ or $Z = 50$

And for $A = 238$, we get $Z = 91.98$ or $Z = 92$

6. For the mirror nuclei $^{23}_{11}\mathrm{Na}_{12}$ and $^{23}_{12}\mathrm{Mg}_{11}$, calculate the Coulomb coefficient a_c and hence estimate r_0. Given:

$M(^{23}\mathrm{Mg}) = 22.994124$ amu

$M(^{23}\mathrm{Na}) = 22.989768$ amu

$M(n) = 1.008665$ amu

$M(p) = 1.007276$ amu

Solution: We have Eq. (2.11)

$$M(A, Z) = ZM_p + (Z-1)\,M_n - a_v A + a_s A^{2/3} + a_c \frac{Z(Z-1)}{A^{1/3}} + a_a \frac{(A-2Z)^2}{A} \mp a_p A^{-3/4}$$

For $^{23}_{12}\mathrm{Mg}_{11}$, $(A - 2Z)^2 = 1$ and for odd A nucleus $a_p = 0$. Taking the approximation $Z(Z-1)$

$\approx Z^2$, this equation reduces to

$$M(A, Z) = ZM_p + (Z-1)\,M_n - a_v A + a_s A^{2/3} + a_c \frac{Z^2}{A^{1/3}} + \frac{a_a}{A}$$

And for $^{23}_{11}\mathrm{Na}_{12}$ nucleus, $(A - 2Z)^2 = 1$ and for odd nuclei $a_p = 0$, we get

$$M(A, Z-1) = (Z-1)\,M_p + ZM_n - a_v A + a_s A^{2/3} + a_c \frac{(Z-1)^2}{A^{1/3}} + \frac{a_a}{A} \mp a_p A^{-3/4}$$

Subtracting these two equations, we get

$$M(A, Z) - M(A, Z-1) = -(M_n - M_p) + \frac{a_c}{A^{1/3}}(2Z-1)$$

Therefore,
$$a_c = \frac{A^{1/3}}{(2Z-1)}\,[M(A, Z) - M(A, Z-1) + (M_n - M_p)]$$

Substituting various values, we get

$$a_c = \frac{23^{1/3}}{(2\times 12-1)}\,[M(^{23}\text{Mg}) - M(^{23}\text{Na}) + (M_n - M_p)]$$

or

$$a_c = \frac{23^{1/3}}{23}\,[22.994124 - 22.989768 + 1.008665 - 1.007276]$$

$$= 0.00071 \text{ amu}$$

Thus,

$$a_c = 0.00071 \times 931.47 = 0.66 \text{ MeV}$$

Now,

$$a_c = -\frac{3}{5}\frac{e^2}{4\pi 4\varepsilon_0 r_0}$$

$$0.66 \times 1.6 \times 10^{-13} = -\frac{3}{5}\frac{8.98\times 10^9 \times (1.6\times 10^{-19})^2}{r_0}$$

which gives

$$r_0 = 1.3 \square 10^{-15} \text{ m}$$

7. Calculate the contribution of Coulomb energy and surface energy terms for $^{236}_{92}$U nucleus.

Solution: Contribution due to Coulomb energy term for $^{236}_{92}$U nucleus

$$= -\frac{3}{5}\frac{Z(Z-1)\,e^2}{4\pi\varepsilon_0 r}$$

$$= -\frac{3}{5}\frac{Z(Z-1)\,e^2}{4\pi\varepsilon_0 r_0 A^{1/3}}$$

Substituting various values, we get

$$= -\frac{3}{5} \times 8.98 \times 10^9 \times \frac{(92 \times 91)\,(1.6 \times 10^{-19})^2}{1.2 \times 10^{-15} \times (236)^{1/3}}$$

$$= -1.557 \times 10^{-10}\ \text{J}$$

$$= -972.0\ \text{MeV}$$

Contribution due to surface energy term for $^{236}_{92}$U nucleus

$$= a_S A^{2/3}$$

$$= -16.8 \square 236^{2/3}$$

$$= -641.6\ \text{MeV}$$

Unsolved Problems

1. Calculate the maximum energy required to remove a neutron from $^{91}_{40}$Zr, $^{92}_{40}$Zr and $^{93}_{40}$Zr. Given:

$M_{\text{H}} = 1.007825$ amu

$M_n = 1.008665$ amu

$M(^{90}\text{Zr}) = 89.904703$ amu

$M(^{91}\text{Zr}) = 90.905644$ amu

$M(^{92}\text{Zr}) = 91.905039$ amu

$M(^{93}\text{Zr}) = 92.906474$ amu [**Ans.** 7.19 MeV, 8.635 MeV and 6.739 MeV]

2. Use liquid drop model to find the most stable isobar for $A = 27$, 64 and 216. [**Ans.** 12, 27, and 73]

3. When thermal neutrons (zero kinetic energy) is captured by a nucleus of Z protons and N neutrons, the excitation energy of the compound nucleus is given by

$$S_n = M(Z, N) + M_n - M(Z, N + 1)$$

Calculate the difference in excitation energy for the case

$$^{235}_{92}\text{U} + n \square\ ^{236}_{92}\text{U} \text{ and } ^{238}_{92}\text{U} + n \square\ ^{239}_{92}\text{U}$$

Given:

$M(^{235}_{92}\text{U}) = 235.04924$ amu

$M(^{236}_{92}\text{U} = 236.045563$ amu

$M(^{238}_{92}\text{U}) = 238.050785$ amu

$M(^{239}_{92}\text{U}) = 239.054290$ amu [**Ans.** 1.74 MeV]

4. Show that average binding energy per nucleon is given as

$$\frac{B}{A} = a_v - \frac{a_s}{A^{1/3}} - a_c \frac{Z^2}{A^{4/3}} - a_a \frac{(A-2Z)^2}{A} \pm a_p A^{-3/4}$$

5. Calculate the binding energy per nucleon in case of ^{124}Xe, using problem 4 and compare with result of Chapter 1. Given:

$m(^{124}\text{Xe}) = 123.9061$ amu [**Ans.** 8.45 MeV, 8.44 MeV]

6. Calculate the surface and Coulomb energy of $^{238}_{92}\text{U}$. [**Ans.** 459.17 MeV, 812.05 MeV]

7. For the mirror nuclei $^{23}_{11}\text{Na}_{12}$ and $^{23}_{12}\text{Mg}_{11}$, calculate the Coulomb term coefficient a_c. Given:

M(Na) = 22.9897 amu

M(Mg) = 22.9941 amu [**Ans.** 0.602 MeV]

8. For the isobaric family with $A = 39$, estimate the nuclear charge of the most stable isobar. [**Ans.** 18]

9. Estimate the separation of the $1p_{1/2}$ and $1d_{5/2}$ energy levels for nuclei with mass number $A \sim 16$. Given the following information:

Total binding energy of ^{15}O = 111.96 MeV

Total binding energy of ^{16}O = 127.62 MeV

Total binding energy of ^{17}O = 131.76 MeV

The sequence of levels is $1s_{1/2}$; $1p_{3/2}$; $1p_{1/2}$; $1d_{5/2}$; $1d_{3/2}$.

[**Ans.** 11.52 MeV]

Section 2.3

Solved Problems

1. Using the shell model, predict the characteristics of ground state of $^{15}_{8}O$, $^{16}_{8}O$ and $^{17}_{8}O$.

Solution: **$^{15}_{8}O$:**

According to shell model, 8 protons pair to give no contribution to the spin. Seven neutrons will be distributed in different shells as

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^1 \mid$$

Last unpaired neutron is in $p_{1/2}$ shell. Therefore, spin of $^{15}_{8}O$ is 1/2. Since the last unpaired neutron is in p shell for which □ = 1, therefore, parity is odd. Therefore, spin parity of $^{15}_{8}O$ is $1^{-}/2$.

$^{16}_{8}O$:

According to shell model, 8 protons pair to give no contribution to the spin. Similarly, 8 neutrons also pair together giving no contribution to spin. Therefore, spin parity of $^{16}_{8}O$ is 0^{+}.

$^{17}_{8}O$:

According to shell model, 8 protons pair to give no contribution to the spin. Nine neutrons will be distributed in different shells as

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^2 \mid (1d_{5/2})^1 (2s_{1/2})^0 (1d_{3/2})^0$$

Last unpaired neutron is in $d_{5/2}$ shell. Therefore, spin of $^{17}_{8}O$ is 5/2. Since the last unpaired neutron is in d shell for which □ = 2, therefore, parity is even. Therefore, spin parity of $^{17}_{8}O$ is $5^{+}/2$.

2. Use single particle shell model to predict the ground state spins and parities of $^{63}_{29}Cu$ and $^{40}_{18}Ar$.

Solution: **$^{63}_{29}Cu_{34}$:**

According to shell model, 34 neutrons pair to give no contribution to the spin. Twenty-nine protons will be distributed in different shells as

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^2 \mid (1d_{5/2})^6 (2s_{1/2})^2 (1d_{3/2})^4 \mid (1f_{7/2})^8 \mid (2p_{3/2})^1 (1f_{5/2})^0$$

Last unpaired proton is in $p_{3/2}$ shell. Therefore, spin of $^{63}_{29}Cu_{34}$ is 3/2. Since the last unpaired proton is in p shell for which □ = 1, therefore, parity is odd. Therefore, spin parity of $^{63}_{29}Cu_{34}$

is $3^-/2$.

$^{40}_{18}Ar_{22}$:

According to shell model, 18 protons pair to give no contribution to the spin. Similarly,
22 neutrons also pair together giving no contribution to spin. Therefore, spin parity of $^{40}_{18}Ar$ is 0^+.

3. Use single particle shell model to predict the ground state spin and parities of $^{19}_{10}Ne$, $^{20}_{10}Ne$ and $^{21}_{10}Ne$.

***Solution:* $^{19}_{10}Ne_9$:**

According to shell model, 10 protons pair to give no contribution to the spin. Nine neutrons will be distributed in different shells as

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^2 \mid (1d_{5/2})^1 (2s_{1/2})^0 (1d_{3/2})^0$$

Last unpaired neutron is in $d_{5/2}$ shell. Therefore, spin of $^{19}_{10}Ne_9$ is 5/2. Since the last unpaired neutron is in d shell for which □ = 2, therefore, parity is even. Therefore, spin parity of $^{19}_{10}Ne_9$ is $5^+/2$.

$^{20}_{10}Ne_{10}$:

According to shell model, 10 protons pair to give no contribution to the spin. Similarly, 10 neutrons also pair together giving no contribution to spin. Therefore, spin parity of $^{20}_{10}Ne$ is 0^+.

$^{21}_{10}Ne_{11}$:

According to shell model, 10 protons pair to give no contribution to the spin. Eleven neutrons will be distributed in different shells as

$$(1s_{1/2})^2 \mid (1p_{3/2})^4 (1p_{1/2})^2 \mid (1d_{5/2})^3 (2s_{1/2})^0 (1d_{3/2})^0$$

Last unpaired neutron is in $d_{5/2}$ shell. Therefore, spin of $^{21}_{10}$Ne is 5/2. Since the last unpaired neutron is in d shell for which □ = 2, therefore, parity is even. Therefore, spin parity of $^{21}_{10}$Ne is $5^+/2$.

4. Calculate the magnetic moments of nuclei in problem 3.

Solution: **$^{19}_{10}$Ne$_9$:**

$$J = \frac{5}{2}$$

The magnetic moment $m = J + 2.29$
or

$$\mu = \frac{5}{2} + 2.29 = 4.79 \text{ nuclear magneton}$$

$^{20}_{10}$Ne$_{10}$:

$$J = 0$$

The magnetic moment $m = J + 2.29$
or

$$m = 2.29 \text{ nuclear magneton}$$

$^{21}_{10}$Ne$_{11}$:

$$J = \tfrac{5}{2}$$

The magnetic moment $m = J + 2.29$
or

$$m = \tfrac{5}{2} + 2.29 = 4.79 \text{ nuclear magneton}$$

Unsolved Problems

1. Predict the shell model state for the odd nucleon in the undermentioned cases:

(i) $^{63}_{29}$Cu

(ii) $^{25}_{12}Mg$

2. Draw schematic shell model diagram for the following nuclei, showing the filling-up of the levels by protons and neutrons.

$$^{7}_{3}Li,\ ^{27}_{13}Al \text{ and } ^{43}_{20}Ca$$

3. The ground state spin of $^{19}_{9}F$ as predicted by the shell model is 5/2, while the experimentally observed spin is 1/2. How can this discrepancy be explained by taking into consideration the order of shell model levels?

4. Predict the characteristics of the ground states of

(i) $^{58}_{27}Co$, $^{59}_{27}Co$ and $^{60}_{27}Co$, and

(ii) $^{207}_{82}Pb$, $^{208}_{82}Pb$ and $^{209}_{82}Pb$

5. Predict the shell model states of the odd nucleon in the following cases:

$^{63}_{29}Cu,\ ^{25}_{12}Mg,\ ^{17}_{9}F,\ ^{127}_{53}I$ $\left[\textbf{Ans.}\ \frac{3^-}{2}, \frac{5^+}{2}, \frac{5^+}{2}, \frac{7^+}{2}\right]$

6. Ground state spin and parity are given for the following nuclei:

$$^{3}He(1/2^+),\ ^{21}Ne(3/2^-) \text{ and } ^{209}Bi(9/2^-)$$

Justify these observed values on the basis of the shell model.

7. Given that the ordering of the nuclear levels is

$$1s_{1/2};\ 1p_{3/2};\ 1p_{1/2};\ 1d_{5/2};\ 1d_{3/2};\ 2s_{1/2};\ 1f_{7/2};\ 2p_{3/2};\ 1f_{5/2};$$

Assign spin and parity to the following nuclei $^{3}_{2}He$, $^{20}_{10}Ne$, $^{27}_{13}Al$, $^{41}_{21}Sc$ and $^{69}_{31}Ga$.

$\left[\textbf{Ans.}\ \frac{1^+}{2}, 0^+, \frac{5^+}{2}, \frac{7^-}{2}, \frac{3^-}{2}\right]$

8. Assume that the level sequence according to shell model is

$$1s_{1/2};\ 1p_{3/2};\ 1p_{1/2};\ 1d_{5/2};\ 2s_{1/2};\ 1d_{3/2};\ 1f_{7/2};\ 2p_{3/2}.$$

Find the ground state spin and parity assignments for the following nuclei:

$^{7}_{3}Li,\ ^{23}_{11}Na$ and $^{33}_{16}S$ $\left[\textbf{Ans.}\ \frac{3^?}{2}, \frac{5^+}{2}, \frac{3^+}{2}\right]$

REVIEW QUESTIONS

Short Answer Type

1. What are the drawbacks of liquid drop model?
2. What are magic numbers?
3. What are the main achievements and limitations of shell model?
4. What are the similarities between a nucleus and liquid drop?
5. Give two achievements of nuclear shell model.
6. What are magic numbers? What is their significance?
7. What are magic numbers, singly magic and doubly magic nuclei? Which nuclear model explains the stability of magic nuclei?
8. What is the importance of asymmetric term in semiempirical mass formula?
9. State predictions of shell model.
10. Name the various contributions to the mass of a nucleus as taken in semiempirical mass formula.
11. Which numbers out of these are magic numbers 6, 8, 12, 14, 20, 40?
12. List various nuclear models.
13. What are magic numbers? Which numbers out of these are magic numbers 6, 8, 10, 12, 14, 16, 18, 20, 25, 30?
14. What was the need of various nuclear models? What are the main successes of liquid drop model?
15. Give two experimental evidences of magic numbers.
16. Discuss various factors in favour of nuclear shell model.

Long Answer Type

1. Discuss the shell model of the nucleus. What are its merits and demerits?
2. Give experimental evidence of magic numbers.
3. Explain the limitations of shell model.
4. Derive the expression for semiempirical mass formula.
5. Discuss the basic assumptions of liquid drop model of nucleus. Explain how the model is used to estimate the semiempirical mass formula.
6. What are assumptions and drawbacks of liquid drop model?

7. What are magic numbers? Give experimental evidences of nuclear magic numbers. What are the achievements of shell model?
8. What do you understand by magic numbers? What are the experimental evidences of nuclear magic numbers?
9. Assuming nucleus to be uniformly charged shell of radius r, calculate the work done against electrostatic force in giving it a charge Ze.
10. Starting from assumptions discuss the achievements and failures of liquid drop model.
11. What is semiempirical mass formula? Explain the significance of various terms.
12. Discuss relative merits and demerits of various nuclear models. What is the importance of magic numbers? Explain shell model in detail.
13. State assumptions of liquid drop model.
14. Derive asymmetric energy term of semiempirical mass formula.
15. Explain the existence of magic numbers on the basis of single particle shell model.
16. Write down the major predictions of shell model.
17. Discuss the stability of nuclei on the basis of semiempirical mass formula.
18. Derive the nuclear energy level scheme on the basis of extreme single particle shell model.
19. List some of the experimental evidences for shell model.
20. Derive the Coulomb energy term of semiempirical mass formula.
21. Give the elementary account of shell model and explain how this model predicts the existence of magic numbers.
22. Give the elementary account of single particle shell model and state its predictions.
23. Explain various predictions of the liquid drop model. Give a brief description of semiempirical mass formula.
24. Derive various factors which contribute to the binding energy of the nucleus and derive semiempirical mass formula based on these factors.
25. State various assumptions of liquid drop model. Discuss the semiempirical mass formula of the liquid drop model. Discuss the

utility of this formula.

26. In what ways liquid drop model is different from the shell model? What are magic numbers? How does the shell model explain the existence of magic numbers and other nuclear properties?

27. What are the basic differences between liquid drop model and shell model of the nucleus? What is the evidence of shell structure in the nuclei? Explain the main assumptions of the shell model of the nucleus. Discuss its achievements, failures and limitations.

28. Explain liquid drop model. Obtain semiempirical mass formula. Give any two achievements of the model.